

NCERT Exemplar Solutions For Grade 12

Mathematics

Chapter 5.- Continuity and Differentiability

Solved Examples

Short Answer Questions

1. Find the value of the constant k so that the function f defined below is continuous at $x = 0$, where

$$f(x) = \begin{cases} \frac{1 - \cos 4x}{8x^2} & \text{at } x \neq 0 \\ k & \text{at } x = 0 \end{cases}$$

Ans: Here, it is given that the function f is continuous at $x = 0$. Therefore, $\lim_{x \rightarrow 0} f(x) = f(0)$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{1 - \cos 4x}{8x^2} = k$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{2\sin^2 2x}{8x^2} = k$$

$$\Rightarrow \lim_{x \rightarrow 0} \left(\frac{\sin 2x}{2x} \right)^2 = k$$

$$\Rightarrow k = 1$$

Hence the value of $k = 1$

2. Discuss the continuity of the function $f(x) = \sin x \cdot \cos x$.

Ans: As we know that $\sin x$ and $\cos x$ are continuous functions and the product of two continuous functions is always a continuous function.

Hence, $f(x) = \sin x \cdot \cos x$ is a continuous function.

3. If $f(x) = \begin{cases} \frac{x^3 + x^2 - 16x + 20}{(x-2)^2} & \text{at } x \neq 2 \\ k & \text{at } x = 2 \end{cases}$

Find the value of k .

Ans: Given $f(2) = k$.

$$\text{Now, } \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2} \frac{x^3 + x^2 - 16x + 20}{(x-2)^2}$$

$$\Rightarrow \lim_{x \rightarrow 2} \frac{(x+5)(x-2)^2}{(x-2)^2}$$

$$\Rightarrow \lim_{x \rightarrow 2} (x+5) = 7$$

Now since f is continuous at $x = 2$, we have

$$\lim_{x \rightarrow 2} f(x) = f(2)$$

$$k = 7$$

4. Show that the function f defined by $f(x) = \begin{cases} x \sin \frac{1}{x} & \text{at } x \neq 0 \\ 0 & \text{at } x = 0 \end{cases}$ is continuous at $x = 0$

Ans: Left hand limit at $x = 0$ is given by

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} x \sin \frac{1}{x} = 0$$

$$\text{Similarly, } \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x \sin \frac{1}{x} = 0$$

$$\text{Also } f(0) = 0$$

$$\text{Hence } \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0)$$

$$\Rightarrow f \text{ is continuous at } x = 0$$

5. Given $f(x) = \frac{1}{x-1}$. Find the points of discontinuity of the composite function $y = f[f(x)]$.

Ans: We know that $f(x) = \frac{1}{x-1}$ is discontinuous at $x = 1$ Now, for $x \neq 1$,

$$f(f(x)) = f\left(\frac{1}{x-1}\right) = \frac{1}{\frac{1}{x-1} - 1} = \frac{x-1}{2-x}$$

which is discontinuous at $x = 2$.

Therefore, the points of discontinuity are $x = 1$ and $x = 2$.

6. Let $f(x) = x|x|$, for all $x \in \mathbb{R}$. Discuss the derivability of $f(x)$ at $x = 0$

Ans: We may rewrite f as $f(x) = \begin{cases} x^2, & \text{if } x \geq 0 \\ -x^2, & \text{if } x \leq 0 \end{cases}$

We know that

$$Lf'(0) = \lim_{x \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{x \rightarrow 0^-} \frac{-h^2 - 0}{h} = \lim_{x \rightarrow 0^-} -h = 0$$

$$Rf'(0) = \lim_{x \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{x \rightarrow 0^+} \frac{h^2 - 0}{h} = \lim_{x \rightarrow 0^+} h = 0$$

Now clearly, the left hand derivative and right hand derivative both are equal, Therefore, f is differentiable at $x = 0$.

7. Differentiate $\sqrt{\tan\sqrt{x}}$ w.r.t x

Ans: Let us consider $y = \sqrt{\tan\sqrt{x}}$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{\tan\sqrt{x}}} \frac{d}{dx} \tan\sqrt{x}$$

$$= \frac{1}{2\sqrt{\tan\sqrt{x}}} \sec^2\sqrt{x} \frac{d}{dx} \sqrt{x}$$

$$= \frac{1}{2\sqrt{\tan\sqrt{x}}} \sec^2\sqrt{x} \frac{1}{2\sqrt{x}}$$

$$= \frac{\sec^2\sqrt{x}}{4\sqrt{x}\sqrt{\tan\sqrt{x}}}$$

8. If $y = \tan(x + y)$, find $\frac{dy}{dx}$.

Ans: Here, $y = \tan(x + y)$.

On differentiating both sides w.r.t. x , we have

$$\frac{dy}{dx} = \frac{d}{dx} \tan(x+y)$$

$$\frac{dy}{dx} = \sec^2(x+y) \frac{d}{dx}(x+y)$$

$$\frac{dy}{dx} = \left(1 + \frac{dy}{dx}\right) \sec^2(x+y)$$

$$\frac{dy}{dx} = \sec^2(x+y) + \frac{dy}{dx} \sec^2(x+y)$$

$$\frac{dy}{dx} - \frac{dy}{dx} \sec^2(x+y) = \sec^2(x+y)$$

$$(1 - \sec^2(x+y)) \frac{dy}{dx} = \sec^2(x+y)$$

$$\frac{dy}{dx} = \frac{\sec^2(x+y)}{1 - \sec^2(x+y)} = -\operatorname{cosec}^2(x+y)$$

Hence, $\frac{dy}{dx} = -\operatorname{cosec}^2(x+y)$

9. If $e^x + e^y = e^{x+y}$, Prove that $\frac{dy}{dx} = -e^{y-x}$.

Ans: Here we have , $e^x + e^y = e^{x+y}$.

On Differentiating both sides w.r.t x , we get

$$\frac{d}{dx} (e^x + e^y) = \frac{d}{dx} e^{x+y}$$

$$e^x + e^y \frac{dy}{dx} = e^{x+y} \frac{d}{dx} (x+y)$$

$$e^x + e^y \frac{dy}{dx} = (1 + \frac{dy}{dx}) e^{x+y}$$

$$e^x + e^y \frac{dy}{dx} = e^{x+y} + e^{x+y} \frac{dy}{dx}$$

$$(e^y - e^{x+y}) \frac{dy}{dx} = e^{x+y} - e^x$$

$$\frac{dy}{dx} = \frac{e^{x+y} - e^x}{e^y - e^{x+y}}$$

$$\frac{dy}{dx} = \frac{e^x + e^y - e^x}{e^y - e^x - e^y}$$

$$\frac{dy}{dx} = -e^{y-x}$$

Hence Proved.

10. Find $\frac{dy}{dx}$ if $y = \tan^{-1} \frac{(3x-x^3)}{1-3x^2}$, $-\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$

Ans: Let us put $x = \tan \theta = \tan^{-1} x$

$$\text{so, } y = \tan^{-1} \frac{(3\tan\theta - \tan^3\theta)}{1-3\tan^2\theta}$$

$$\text{We know that } \tan 3\theta = \frac{(3\tan\theta - \tan^3\theta)}{1-3\tan^2\theta}$$

$$y = \tan^{-1} \tan 3\theta$$

$$y = 3\theta \left(-\frac{\pi}{2} < 3\theta < \frac{\pi}{2} \right)$$

$$y = 3 \tan^{-1} x$$

Now on differentiating both sides w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}(3 \tan^{-1} x)$$

$$\frac{dy}{dx} = 3 \frac{1}{1+x^2}$$

Hence, $\frac{dy}{dx} = \frac{3}{1+x^2}$

11. If $y = \sin^{-1} \{x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}\}$ and $0 < x < 1$, then find $\frac{dy}{dx}$.

Ans: we have

$$y = \sin^{-1} \{x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}\}; 0 < x < 1$$

Now, let's put $x = \sin A$ and $\sqrt{x} = \sin B$

$$y = \sin^{-1} \{ \sin A \sqrt{1 - \sin^2 B} - \sin B \sqrt{1 - \sin^2 A} \}$$

$$= \sin^{-1} \{ \sin A \cos B - \sin B \cos A \}$$

$$= \sin^{-1} \{ \sin(A - B) \} = A - B$$

$$y = \sin^{-1} x - \sin^{-1} \sqrt{x}$$

On Differentiating w.r.t to x, we get

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-\sqrt{x^2}}} \cdot \frac{d}{dx}(\sqrt{x})$$

$$= \frac{1}{\sqrt{1-x^2}} - \frac{1}{2\sqrt{x} \cdot \sqrt{1-x}}$$

Therefore, $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} - \frac{1}{2\sqrt{x} \cdot \sqrt{1-x}}$

12. If $x = a \sec^3 \theta$ and $y = a \tan^3 \theta$, find $\frac{dy}{dx}$ at $\theta = \frac{\pi}{3}$

Ans: we have $x = a \sec^3 \theta$ and $y = a \tan^3 \theta$

On Differentiating both sides w.r.t θ

$$\frac{dx}{d\theta} = 3a \sec^2 \theta \sec \theta \cdot \tan \theta$$

$$\text{and } \frac{dy}{d\theta} = 3a \tan^2 \theta \sec^2 \theta$$

$$\text{Now, } \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{3a \tan^2 \theta \sec^2 \theta}{3a \sec^2 \theta \sec \theta \cdot \tan \theta} = \sin \theta$$

$$\text{At } \theta = \frac{\pi}{3}$$

$$\frac{dy}{dx} = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

Hence,

$$\frac{dy}{dx} = \frac{\sqrt{3}}{2}, \text{ at } \theta = \frac{\pi}{3}$$

13. If $x^y = e^{(x-y)}$, Prove that $\frac{dy}{dx} = \frac{\log x}{(1 + \log x)^2}$

Ans: we have,

$$x^y = e^{(x-y)}$$

Taking log on both the sides we get ,

$$\log(x^y) = \log(e^{x-y})$$

$$y \log x = (x-y) \log e$$

$$y \log x = (x-y) \cdot 1$$

$$y (\log x + 1) = x$$

$$y = \frac{x}{1 + \log x}$$

Using Quotient Rule, we get

$$\frac{dy}{dx} = \frac{(1 + \log x) \cdot \frac{d}{dx}(x) - x \cdot \frac{d}{dx}(1 + \log x)}{(1 + \log x)^2}$$

$$\frac{dy}{dx} = \frac{(1 + \log x) - 1}{(1 + \log x)^2}$$

$$\frac{dy}{dx} = \frac{\log x}{(1 + \log x)^2}$$

Hence Proved.

14. If $y = \tan x + \sec x$, prove that $\frac{d^2 y}{dx^2} = \frac{\cos x}{(1 - \sin x)^2}$

Ans: we have,

$$y = \tan x + \sec x$$

On differentiating both sides w.r.t x , we get

$$\frac{dy}{dx} = \frac{d}{dx}(\tan x) + \frac{d}{dx}(\sec x)$$

$$\frac{dy}{dx} = \sec^2 x + \sec x \cdot \tan x$$

$$\frac{dy}{dx} = \frac{1}{\cos^2 x} + \frac{\sin x}{\cos^2 x} = \frac{1 + \sin x}{\cos^2 x}$$

$$\frac{dy}{dx} = \frac{1 + \sin x}{\cos^2 x}$$

$$\frac{dy}{dx} = \frac{1 + \sin x}{1 - \sin^2 x}$$

$$\frac{dy}{dx} = \frac{1 + \sin x}{(1 + \sin x)(1 - \sin x)}$$

$$\frac{dy}{dx} = \frac{1}{(1 - \sin x)}$$

Again Differentiating both sides w.r.t x

$$\frac{d^2 y}{dx^2} = \frac{-(-\cos x)}{(1 - \sin x)^2}$$

$$\frac{d^2 y}{dx^2} = \frac{\cos x}{(1 - \sin x)^2}$$

Hence Proved .

15.If $f(x) = |\cos x|$, find $f'(\frac{3\pi}{4})$

Ans: we know that $\cos x < 0$ when $\frac{\pi}{2} < x < \pi$

$$|\cos x| = -\cos x$$

$$\text{so, } f(x) = -\cos x$$

$$f'(x) = \sin x$$

$$\text{Therefore, } f'\left(\frac{3\pi}{4}\right) = \sin \frac{3\pi}{4}$$

$$f'\left(\frac{3\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

Hence Proved

16. If $f(x) = |\cos x - \sin x|$, find $f'(\frac{\pi}{6})$

Ans: we know that $\cos x > \sin x$ when $0 < x < \frac{\pi}{4}$

$$\cos x - \sin x > 0$$

$$\text{so, } f(x) = \cos x - \sin x$$

$$f'(x) = -\sin x - \cos x$$

$$\text{Therefore, } f'(\frac{\pi}{6}) = -\sin \frac{\pi}{6} - \cos \frac{\pi}{6}$$

$$f'(\frac{\pi}{6}) = -\frac{1}{2}(1 + \sqrt{3})$$

Hence Proved

17. Verify Rolle's theorem for the function, $f(x) = \sin 2x$ in $[0, \frac{\pi}{2}]$

Ans: we have,

$$f(x) = \sin 2x \text{ in } [0, \frac{\pi}{2}]$$

The given function is continuous in $[0, \frac{\pi}{2}]$ as f is a sine function.

$f'(x) = 2\cos 2x$, exists in $(0, \frac{\pi}{2})$, hence f is differentiable in $(0, \frac{\pi}{2})$

$$f(0) = \sin 0 = 0 \text{ and } f(\frac{\pi}{2}) = \sin \frac{\pi}{2} = 0$$

$$f(0) = f(\frac{\pi}{2})$$

Since, all the conditions of Rolle's theorem are satisfied, so there exists a $c \in$

$$\left(0, \frac{\pi}{2}\right) \text{ such that } f'(c) = 0$$

$$f'(x) = 2\cos 2x$$

$$f'(c) = 2\cos 2c = 0$$

$$\cos 2c = 0$$

$$2c = \frac{\pi}{2}$$

$$c = \frac{\pi}{4}$$

18. Verify mean value theorem for the function $f(x) = (x - 3)(x - 6)(x - 9)$ in $[3, 5]$.

Ans:(i) Given function $f(x)$ is continuous in $[3, 5]$ as the product of polynomial functions is a polynomial and all polynomial functions are continuous

(ii) $f'(x) = 3x^2 - 36x + 99$ exists in $(3, 5)$ and hence is derivable in $(3, 5)$.

Thus conditions of the mean value theorem are satisfied. Hence, there exists at least one $c \in (3, 5)$ such that

$$f'(c) = \frac{f(5) - f(3)}{5 - 3}$$

$$3c^2 - 36c + 99 = \frac{8 - 0}{2}$$

$$3c^2 - 36c + 99 = 4$$

$$3c^2 - 36c + 95 = 0$$

$$c = 6 \pm \sqrt{\frac{13}{3}}$$

$$c \neq 6 + \sqrt{\frac{13}{3}}$$

Therefore, $c = 6 - \sqrt{\frac{13}{3}}$

Long Answers(L.A.)

19. If $f(x) = \frac{\sqrt{2}\cos x - 1}{\cot x - 1}$, $x \neq \frac{\pi}{4}$, Find the value of $f\left(\frac{\pi}{4}\right)$ so that $f(x)$ becomes continuous at $x = \frac{\pi}{4}$

Ans: we have

$$f(x) = \frac{\sqrt{2}\cos x - 1}{\cot x - 1}, x \neq \frac{\pi}{4}$$

$$\lim_{x \rightarrow \frac{\pi}{4}} f(x) = \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sqrt{2}\cos x - 1}{\cot x - 1}$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{(\sqrt{2}\cos x - 1)\sin x}{\cos x - \sin x}$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{(\sqrt{2}\cos x - 1) \frac{(\sqrt{2}\cos x + 1)(\cos x + \sin x)\sin x}{(\sqrt{2}\cos x + 1)(\cos x + \sin x)}}{\cos x - \sin x}$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{(2\cos^2 x - 1)(\cos x + \sin x)}{(\cos^2 x - \sin^2 x)(\sqrt{2}\cos x + 1)} \cdot \sin x$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos 2x (\cos x + \sin x)}{\cos 2x (\sqrt{2}\cos x + 1)} \cdot \sin x$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{(\cos x + \sin x)}{(\sqrt{2}\cos x + 1)} \cdot \sin x$$

$$= \frac{\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right)}{\frac{1}{\sqrt{2}} \cdot \sqrt{2} + 1} \times \frac{1}{\sqrt{2}} = \frac{1}{2}$$

Therefore, $\lim_{x \rightarrow \frac{\pi}{4}} f(x) = \frac{1}{2}$

Hence, for f to be continuous at $x = \frac{\pi}{4}$, $f\left(\frac{\pi}{4}\right)$ should be equal to $\frac{1}{2}$

i.e. $f\left(\frac{\pi}{4}\right) = \frac{1}{2}$

20. Show that the function f given by $f(x) = \begin{cases} \frac{e^{\frac{1}{x}} - 1}{e^{\frac{1}{x}} + 1}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ is discontinuous at $x = 0$.

discontinuous at $x = 0$.

Ans: The left hand limit of f at $x = 0$ is given by

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{e^{\frac{1}{x}} - 1}{e^{\frac{1}{x}} + 1} = \frac{0 - 1}{0 + 1} = -1$$

Similarly,

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{e^{\frac{1}{x}} - 1}{e^{\frac{1}{x}} + 1} = \frac{1 - 0}{1 + 0} = 1$$

Therefore,

$$\lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x), \text{ hence } \lim_{x \rightarrow 0} f(x) \text{ does not exist.}$$

f is discontinuous at $x = 0$

21. Let $f(x) = \begin{cases} \frac{1-\cos 4x}{x^2}, & \text{if } x < 0 \\ a, & \text{if } x = 0 \\ \frac{\sqrt{x}}{\sqrt{16+\sqrt{x}}-4}, & \text{if } x > 0 \end{cases}$

For what value of a , f is continuous at $x = 0$?

Ans: Here $f(0) = a$ Left hand limit of f at 0 is

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{1-\cos 4x}{x^2}$$

$$= \lim_{x \rightarrow 0^-} \frac{2\sin^2 2x}{x^2}$$

$$= \lim_{x \rightarrow 0^-} 8 \left(\frac{\sin 2x}{x} \right)^2$$

$$= 8 \cdot (1)^2$$

$$= 8$$

$$\text{Now, } \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{\sqrt{16+\sqrt{x}}-4}$$

$$= \lim_{x \rightarrow 0^+} \frac{\sqrt{x}(\sqrt{16+\sqrt{x}}+4)}{(\sqrt{16+\sqrt{x}}-4)(\sqrt{16+\sqrt{x}}+4)}$$

$$= \lim_{x \rightarrow 0^+} \frac{\sqrt{x}(\sqrt{16+\sqrt{x}}+4)}{(\sqrt{16+\sqrt{x}}-4)(\sqrt{16+\sqrt{x}}+4)}$$

$$= \lim_{x \rightarrow 0^+} \frac{\sqrt{x}(\sqrt{16+\sqrt{x}}+4)}{(16+\sqrt{x}-16)}$$

$$= \lim_{x \rightarrow 0^+} \frac{\sqrt{x}(\sqrt{16+\sqrt{x}}+4)}{\sqrt{x}}$$

$$= \lim_{x \rightarrow 0^+} (\sqrt{16+\sqrt{x}}+4)$$

$$= 8$$

Therefore, $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = 8$, hence f is continuous at $x = 0$ only if $a = 8$.

22. Examine the differentiability of the function f defined by

$$f(x) = \{ 2x+3, \text{if } -3 \leq x < -2$$

$$\{ x+1, \text{if } -2 \leq x < 0$$

$$\{ x+2, \text{if } 0 \leq x \leq 1$$

Ans: The points where differentiability can't be directly predicted of $f(x)$ are $x = -2$ and $x = 0$.

Let us find Differentiability at $x = -2$ and at $x = 0$

Differentiability at $x = -2$

$$\text{L.H.D} = f'(-2) = \lim_{h \rightarrow 0^-} \frac{f(-2+h) - f(-2)}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{2(-2+h)+3 - (-2+1)}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{2h}{h}$$

$$= 2$$

$$\text{R.H.D} = f'(-2) = \lim_{h \rightarrow 0^+} \frac{f(-2+h) - f(-2)}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{-2+h+1 - (-2+1)}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{h}{h}$$

$$= 1$$

Clearly, $\text{L.H.D} \neq \text{R.H.D}$

$f(x)$ is not differentiable at $x = -2$

Now, Differentiability at $x = 0$

$$\text{L.H.D} = f'(0) = \lim_{h \rightarrow 0^-} \frac{f(-2+h) - f(-2)}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{(0+h+1) - (0+2)}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{h-1}{h}$$

$$= \lim_{h \rightarrow 0^-} \left(1 - \frac{1}{h}\right)$$

which does not exist. Hence f is not differentiable at $x = 0$.

Therefore, $f(x)$ is not differentiable at points $x = -2$ and $x = 0$

23. Differentiate $u = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$ w.r.t.v = $\cos^{-1}(2x\sqrt{1-x^2})$, where $x \in \left(\frac{1}{\sqrt{2}}, 1\right)$

Ans: Given,

$$u = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right) \text{ and } v = \cos^{-1}(2x\sqrt{1-x^2})$$

$$\text{we have, } u = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$$

$$\text{Put } x = \sin \theta, \text{ where } \frac{\pi}{4} < \theta < \frac{\pi}{2}$$

$$u = \tan^{-1}\left(\frac{\sqrt{1-\sin^2\theta}}{\sin\theta}\right)$$

$$u = \tan^{-1} \left(\frac{\cos \theta}{\sin \theta} \right)$$

$$u = \tan^{-1} (\cot \theta)$$

$$u = \tan^{-1} \left(\tan \left(\frac{\pi}{2} - \theta \right) \right)$$

$$u = \left(\frac{\pi}{2} - \theta \right)$$

$$u = \left(\frac{\pi}{2} - \sin^{-1} x \right)$$

On differentiating both sides w.r.t x

$$\frac{du}{dx} = \frac{-1}{\sqrt{1-x^2}} \dots \dots \dots \text{eqn (1)}$$

$$\text{Now, } v = \cos^{-1} (2x\sqrt{1-x^2})$$

$$v = \frac{\pi}{2} - \sin^{-1} (2x\sqrt{1-x^2})$$

Again, Put $x = \sin \theta$, where $\frac{\pi}{4} < \theta < \frac{\pi}{2}$

$$v = \frac{\pi}{2} - \sin^{-1} (2\sin \theta \sqrt{1-\sin^2 \theta})$$

$$v = \frac{\pi}{2} - \sin^{-1} (2\sin \theta \cos \theta)$$

$$v = \sin^{-1} (\sin 2\theta)$$

$$v = 2\theta$$

$$v = 2\sin^{-1} x$$

On differentiating both sides w.r.tx

$$\frac{dv}{dx} = \frac{2}{\sqrt{1-x^2}} \dots \dots \dots \text{eqn(2)}$$

$$\text{Therefore, } \frac{du}{dv} = \frac{\frac{dx}{dv}}{\frac{dx}{dx}} = \frac{\frac{-1}{2\sqrt{1-x^2}}}{1} \dots \text{from eqn(1) \&eqn(2)}$$

$$\frac{du}{dv} = \frac{-1}{2}$$

Objective Type Questions

24. The function $f(x) = \begin{cases} \frac{\sin x}{x} + \cos x, & \text{if } x \neq 0 \\ k, & \text{if } x = 0 \end{cases}$ is continuous at $x = 0$, then the value of k is

(A) 3

(B) 2

(C) 1

(D) 1.5

Ans: Since the function $f(x)$ is continuous at $x = 0$

$$\text{L.H.L} = \text{R.H.L} = f(0)$$

$$\lim_{x \rightarrow 0^+} \left(\frac{\sin x}{x} + \cos x \right) = f(0)$$

Put $x = 0 + h$

$$\lim_{h \rightarrow 0} \left(\frac{\sin h}{h} + \cos h \right) = k$$

Using identities : $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ and $\lim_{x \rightarrow 0} \cos x = 1$

$$1 + 1 = k$$

$$k = 2$$

Therefore, the value of $k = 2$

Hence, the correct option is (B)

25. The function $f(x) = [x]$, where $[x]$ denotes the greatest integer function, is continuous at

(A) 4

(B) -2

(C) 1

(D) 1.5

Ans: The greatest integer function $f(x) = [x]$ is discontinuous at all integral values of x .

Hence the correct option is (D)

26. The number of points at which the function $f(x) = \frac{1}{x - [x]}$ is not continuous is

(A) 1

(B) 2

(C) 3

(D) none of these

Ans: As we know that $x - [x] = 0$, when x is an integer therefore $f(x)$ is discontinuous for all $x \in \mathbb{Z}$.

Hence, the correct option is (D)

27. The function given by $f(x) = \tan x$ is discontinuous on the set

(A) $\{n\pi : n \in \mathbb{Z}\}$

(B) $\{2n\pi : n \in \mathbb{Z}\}$

(C) $\left\{(2n+1)\frac{\pi}{2} : n \in \mathbb{Z}\right\}$

(D) $\left\{\frac{n\pi}{2}:n \in \mathbb{Z}\right\}$

Ans: $f(x) = \tan x$ is always discontinuous on the set $\left\{(2n+1)\frac{\pi}{2}:n \in \mathbb{Z}\right\}$

Hence the correct option is (C)

28. Let $f(x) = |\cos x|$. Then,

(A) f is everywhere differentiable.

(B) f is everywhere continuous but not differentiable at $x = n\pi, n \in \mathbb{Z}$.

(C) f is everywhere continuous but not differentiable at $x = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$

(D) none of these

Ans: We have

$$f(x) = |\cos x|$$

Here, f is everywhere continuous but not differentiable at $x = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$

Hence the correct option is (C)

29. The function $f(x) = |x| + |x - 1|$ is

(A) continuous at $x = 0$ as well as at $x = 1$.

(B) continuous at $x = 1$ but not at $x = 0$.

(C) discontinuous at $x = 0$ as well as at $x = 1$.

(D) continuous at $x = 0$ but not at $x = 1$.

Ans: $f(x) = |x| + |x - 1|$ is continuous at $x = 0$ as well as at $x = 1$.

Hence the correct option is (A)

30. The value of k which makes the function defined by $f(x) = \begin{cases} \sin \frac{1}{x}, & \text{if } x \neq 0 \\ k, & \text{if } x = 0 \end{cases}$ continuous at $x=0$ is

- (A) 8
- (B) 1
- (C) -1
- (D) none of these

Ans: As we know that $\lim_{x \rightarrow 0} \sin \frac{1}{x}$ does not exist

Hence the correct option is (D)

31. The set of points where the functions f given by $f(x) = |x - 3| \cos x$ is differentiable is

- (A) $\mathbb{R}$
- (B) $\mathbb{R} - \{3\}$
- (C) $(0, \infty)$
- (D) none of these

Ans: The set of points where the functions f given by $f(x) = |x - 3| \cos x$ is differentiable is $\mathbb{R} - \{3\}$

Hence the correct option is (B)

32. Differential coefficient of $\sec(\tan^{-1} x)$ w.r.t. x is

- (A) $\frac{x}{\sqrt{1+x^2}}$
- (B) $\frac{x}{1+x^2}$
- (C) $x\sqrt{1+x^2}$

(D) $\frac{1}{1+x^2}$

Ans: Let $y = \sec(\tan^{-1}x)$

On Differentiating both sides w.r.t x

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \sec(\tan^{-1}x) \\ &= \sec(\tan^{-1}x) \cdot \tan(\tan^{-1}x) \cdot \frac{d}{dx}(\tan^{-1}x) \\ &= x \cdot \sec(\tan^{-1}x) \cdot \frac{1}{1+x^2}\end{aligned}$$

Clearly the differential coefficient of $\sec(\tan^{-1}x)$ is $\frac{x}{1+x^2}$

Hence the correct option is (B)

33. If $u = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ and $v = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$, then $\frac{du}{dv}$ is

(A) $\frac{1}{2}$

(B) x

(C) $\frac{1-x^2}{1+x^2}$

(D) 1

Ans: Given $u = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ and $v = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$

we have ,

$$u = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$$

Put $x = \sin\theta$

$$u = \sin^{-1}\left(\frac{2\sin\theta}{1+\sin^2\theta}\right)$$

$$u = \sin^{-1}(\sin 2\theta)$$

$$u = 2\theta$$

$$\frac{du}{d\theta} = 2 \dots \text{eqn (1)}$$

Now, we have ,

$$v = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$$

$$\text{put } v = \tan \theta$$

$$v = \tan^{-1}\left(\frac{2\tan\theta}{1-\tan^2\theta}\right)$$

$$v = \tan^{-1}(\tan 2\theta)$$

$$v = 2\theta$$

$$\frac{dv}{d\theta} = 2 \dots \dots \text{eqn(2)}$$

Dividing equation (1) by(2)

$$\frac{\frac{du}{d\theta}}{\frac{dv}{d\theta}} = \frac{2}{2} = 1$$

$$\frac{du}{dv} = 1$$

Hence the correct option is (D)

34. The value of c in Rolle's Theorem for the function $f(x) = e^x \sin x$, $x \in \pi [0, \pi]$ is

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{2}$

(D) $\frac{3\pi}{4}$

Ans:

We Know that by Rolle's Theorem that a function $f(x)$ is continuous on $[a,b]$ and differentiable in (a,b) then there exists $c \in (a,b)$ such that $f'(c) = 0$

We have ,

$$f(x) = e^x \sin x, x \in \pi [0, \pi] \text{ is}$$

$$f'(x) = e^x \sin x + e^x \cos x$$

$$f'(c) = e^c (\sin c + \cos c) = 0$$

Now we know that $e^c \neq 0$

$$\tan c = -1$$

$$c = \frac{3\pi}{4}$$

Hence option (D) is the correct answer.

35. The value of c in Mean value theorem for the function $f(x) = x(x - 2)$, $x \in [1, 2]$ is

(A) $\frac{3}{2}$

(B) $\frac{2}{3}$

(C) $\frac{1}{2}$

(D) $\frac{3}{2}$

Ans: We have,

$$f(x) = x(x - 2)$$

$$f(x) = x^2 - 2x$$

Since the polynomial function is always continuous and differentiable

$f(x) = x^2 - 2x$ is always continuous on $[1, 2]$ and differentiable in $(1, 2)$

Now since, $f(x)$ satisfies both the conditions of Mean Value Theorem there exist a real number $c \in (1, 2)$ such that

$$f'(c) = \left(\frac{f(2) - f(1)}{2 - 1} \right)$$

$$f'(c) = \left(\frac{f(2) - f(1)}{1} \right)$$

$$f'(c) = f(2) - f(1)$$

$$f'(c) = f(2) - f(1)$$

we have $f(x) = x^2 - 2x$

$$f'(x) = 2x - 2$$

$$f'(c) = 2c - 2$$

$$f(2) - f(1) = 2c - 2$$

$$2^2 - 2 \times 2 - (1^2 - 2 \times 1) = 2c - 2$$

$$0 - (-1) = 2c - 2$$

$$2c = 1 + 2$$

$$c = \frac{3}{2}$$

Clearly $c = \frac{3}{2} \in (1, 2)$

Hence option (A) is the correct answer

36. Match the following Column-I Column-II

(A) If a function $f(x) = \begin{cases} \frac{\sin 3x}{x}, & \text{if } x \neq 0 \\ \frac{k}{2}, & \text{if } x = 0 \end{cases}$ is continuous at $x = 0$, then k is equal to

(B) Every continuous function is differentiable (b) True (C) An example of a function which is continuous (c) everywhere but not differentiable at exactly one point

(D) The identity function i.e. $f(x) = x \forall x \in \mathbb{R}$ is ad) False

Ans: $A \rightarrow c, B \rightarrow d, C \rightarrow a, D \rightarrow b$

Fill in the blanks in each of the Examples 37 to 41

37. The number of points at which the function $f(x) = \frac{1}{\log|x|}$ is discontinuous is _____.

Ans: The given function is discontinuous at $x = 0, \pm 1$ and hence the number of points of discontinuity is 3.

Therefore, The number of points at which the function $f(x) = \frac{1}{\log|x|}$ is discontinuous is 3

38. If $f(x) = \begin{cases} ax+1, & x \geq 1 \\ x+2, & x < 1 \end{cases}$ is continuous, then a should be equal to _____.

Ans: If $f(x) = \begin{cases} ax+1, & x \geq 1 \\ x+2, & x < 1 \end{cases}$ is continuous, then a should be equal to 2.

39. The derivative of ${}^{\log}10^x$ w.r.t. x is _____.

Ans: The derivative of ${}^{\log}10^x$ w.r.t. x is $({}^{\log}10^e) \cdot \frac{1}{x}$

40. If $y = \sec^{-1} \frac{\sqrt{x+1}}{\sqrt{x-1}} + \sin^{-1} \frac{\sqrt{x-1}}{\sqrt{x+1}}$, then $\frac{dy}{dx}$ is equal to _____

Ans: $y = \sec^{-1} \frac{\sqrt{x+1}}{\sqrt{x-1}} + \sin^{-1} \frac{\sqrt{x-1}}{\sqrt{x+1}}$, then $\frac{dy}{dx}$ is equal to 0

41. The derivative of $\sin x$ w.r.t. $\cos x$ is _____

Ans: The derivative of $\sin x$ w.r.t. $\cos x$ is $-\cot x$

State whether the statements are True or False in each of the Exercises 42 to 46.

42. For continuity, at $x = a$, each of $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ is equal to $f(a)$.

Ans: We know that for a continuous function at $x = a$

$$L.H.L = R.H.L = f(a)$$

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$$

Hence the given statement is True.

43. $y = |x - 1|$ is a continuous function .

Ans: Yes, $y = |x - 1|$ is a continuous function .

Hence the given statement is True.

44. A continuous function can have some points where the limit does not exist.

Ans: A continuous function can never have some points where the limit does not exist.

Hence the given statement is False .

45. $|\sin x|$ is a differentiable function for every value of x .

Ans: $|\sin x|$ is not a differentiable function for every value of x .

Hence the given statement is False

46. $\cos|x|$ is differentiable everywhere.

Ans: Yes, $\cos|x|$ is differentiable everywhere.

Hence the given statement is True.

Exercise

Short Answer Type Questions

1. Examine the continuity of the function $f(x) = x^3 + 2x^2 - 1$ at $x = 1$.

Ans: Given that $f(x) = x^3 + 2x^2 - 1$ we must check continuity at $x = 1$.

$$RHL = f(x)$$

$$= \lim_{x \rightarrow 1^+} (x^3 + 2x^2 - 1) ; \text{ here, } x = 1 + h, h \rightarrow 0$$

$$= \lim_{h \rightarrow 0} (1 + h)^3 + 2(1 + h)^2 - 1$$

$$= 1 + 2 - 1$$

$$RHL = 2$$

$$LHL = f(x)$$

$$= \lim_{x \rightarrow 1^-} (x^3 + 2x^2 - 1) ; \text{ here, } x = 1 - h, h \rightarrow 0$$

$$= \lim_{h \rightarrow 0} (1 - h)^3 + 2(1 - h)^2 - 1$$

$$= 1 + 2 - 1$$

$$LHL = 2$$

$$f(1) = 1^3 + 2 \times 1^2 - 1 = 2$$

So, we have $LHL = RHL = f(x)$ at $x = 1$.

Hence $f(x)$ is continuous at $x = 1$.

Find which of the functions in Exercises 2 to 10 is continuous or discontinuous at the indicated points:

2. $f(x) = \begin{cases} 3x + 5, & \text{if } x \geq 2 \\ x^2, & \text{if } x < 2 \end{cases}$ at $x = 2$.

Ans: Given that, $f(x) = \begin{cases} 3x + 5, & \text{if } x \geq 2 \\ x^2, & \text{if } x < 2 \end{cases}$ at $x = 2$

$$LHL = f(x) = \lim_{x \rightarrow 2^-} x^2$$

$$= 2^2$$

$$LHL = 4$$

$$RHL = f(x) = \lim_{x \rightarrow 2^+} (3x + 5)$$

$$= 3 \times 2 + 5$$

$$RHL = 11$$

$$f(2) = 3 \times 2 + 5 = 11$$

Here $LHL \neq RHL = f(x)$ at $x = 2$

Hence $f(x)$ is discontinuous at $x = 2$.

3. $f(x) = \begin{cases} \frac{1 - \cos 2x}{x^2}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ at $x = 0$.

Ans: Given that, $f(x) = \begin{cases} \frac{1 - \cos 2x}{x^2}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ at $x = 0$

$$LHL = f(x)$$

$$= \lim_{x \rightarrow 0^-} \frac{1 - \cos 2x}{x^2}$$

$$= \lim_{x \rightarrow 0^-} \frac{2x}{x^2}$$

$$= 2 \times (1)^2$$

$$LHL = 2$$

$$RHL = f(x)$$

$$= \lim_{x \rightarrow 0^+} \frac{1 - \cos 2x}{x^2}$$

$$= \lim_{x \rightarrow 0^+} \frac{2x}{x^2}$$

$$= 2 \times (1)^2$$

$$RHL = 2$$

$$f(0) = 5$$

Here $LHL = RHL \neq f(x)$ at $x = 0$

Hence $f(x)$ is discontinuous at $x = 0$.

$$4. f(x) = \begin{cases} \frac{2x^2 - 3x - 2}{x - 2}, & \text{if } x \neq 2 \\ 5, & \text{if } x = 2 \end{cases} \text{ at } x = 2.$$

$$\text{Ans: Given that, } f(x) = \begin{cases} \frac{2x^2 - 3x - 2}{x - 2}, & \text{if } x \neq 2 \\ 5, & \text{if } x = 2 \end{cases} \text{ at } x = 2$$

$$LHL = f(x)$$

$$= \lim_{x \rightarrow 2^-} \frac{2x^2 - 3x - 2}{x - 2}$$

$$= \lim_{x \rightarrow 2^-} \frac{(2x + 1)(x - 2)}{x - 2}$$

$$= \lim_{x \rightarrow 2^-} (2x + 1)$$

$$= 2 \times 2 + 1$$

$$LHL = 5$$

$$RHL = f(x)$$

$$= \lim_{x \rightarrow 2^-} \frac{2x^2 - 3x - 2}{x - 2}$$

$$= \lim_{x \rightarrow 2^-} \frac{(2x+1)(x-2)}{x-2}$$

$$= \lim_{x \rightarrow 2^-} (2x+1)$$

$$= 2 \times 2 + 1$$

$$RHL = 5$$

$$f(2) = 5$$

Here $LHL = RHL = f(x)$ at $x = 2$

Hence $f(x)$ is continuous at $x = 2$.

$$5. f(x) = \begin{cases} \frac{|x-4|}{2(x-4)}, & \text{if } x \neq 4 \\ 0, & \text{if } x = 4 \end{cases} \text{ at } x = 4.$$

$$\text{Ans: Given that, } f(x) = \begin{cases} \frac{|x-4|}{2(x-4)}, & \text{if } x \neq 4 \\ 0, & \text{if } x = 4 \end{cases} \text{ at } x = 4$$

We know that $|x-4| = \begin{cases} -(x-4), & x < 4 \\ (x-4), & x > 4 \end{cases}$

$$LHL = f(x)$$

$$= \lim_{x \rightarrow 4^-} \frac{|x-4|}{2(x-4)}$$

$$= \lim_{x \rightarrow 4^-} \frac{-(x-4)}{2(x-4)}$$

$$LHL = -\frac{1}{2}$$

$$RHL = f(x)$$

$$= \lim_{x \rightarrow 4^+} \frac{|x-4|}{2(x-4)}$$

$$= \lim_{x \rightarrow 4^+} \frac{(x-4)}{2(x-4)}$$

$$RHL = \frac{1}{2}$$

$$f(4) = 0$$

Here $LHL \neq RHL \neq f(x)$ at $x = 4$

Hence $f(x)$ is discontinuous at $x = 4$.

$$6. f(x) = \begin{cases} |x| \cos\left(\frac{1}{x}\right), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases} \text{ at } x = 0.$$

Ans: Given that $f(x) = \begin{cases} |x| \cos\left(\frac{1}{x}\right), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ at $x = 0$.

$$LHL = f(x)$$

$$= \lim_{x \rightarrow 0^-} |x| \cos\left(\frac{1}{x}\right)$$

$$= \lim_{x \rightarrow 0^-} (-x) \cos\left(\frac{1}{x}\right)$$

$$= 0 \times (\text{number which will vary from } -1 \text{ to } 1)$$

$$LHL = 0$$

$$RHL = f(x)$$

$$= \lim_{x \rightarrow 0^+} |x| \cos\left(\frac{1}{x}\right)$$

$$= \lim_{x \rightarrow 0^+} (x) \cos \cos \left(\frac{1}{x} \right)$$

$$= 0 \times (\text{number which will vary from } -1 \text{ to } 1)$$

$$RHL = 0$$

$$f(0) = 0$$

$$\text{Here } LHL = RHL = f(x) \text{ at } x = 0$$

Hence $f(x)$ is continuous at $x = 0$.

$$7. f(x) = \begin{cases} |x-a| \sin \sin \left(\frac{1}{x-a} \right), & \text{if } x \neq a \\ 0, & \text{if } x = a \end{cases} \text{ at } x = a.$$

Ans:

$$\text{Given that } f(x) = \begin{cases} |x-a| \sin \sin \left(\frac{1}{x-a} \right), & \text{if } x \neq a \\ 0, & \text{if } x = a \end{cases} \text{ at } x = a.$$

$$LHL = f(x)$$

$$= \lim_{x \rightarrow a^-} |x-a| \sin \sin \left(\frac{1}{x-a} \right);$$

$$A. |x-a| = \begin{cases} -(x-a), & x < a \\ x-a, & x > a \end{cases}$$

$$= \lim_{x \rightarrow a^-} (-(x-a)) \sin \sin \left(\frac{1}{x-a} \right)$$

$$= 0 \times (\text{number which will vary from } -1 \text{ to } 1)$$

$$LHL = 0$$

$$RHL = f(x)$$

$$= \lim_{x \rightarrow a^+} |x-a| \sin \sin \left(\frac{1}{x-a} \right);$$

$$B. |x-a| = \begin{cases} -(x-a), & x < a \\ x-a, & x > a \end{cases}$$

$$= \lim_{x \rightarrow a^+} ((x-a)) \sin \left(\frac{1}{x-a} \right)$$

$$= 0 \times (\text{number which will vary from } -1 \text{ to } 1)$$

$$RHL = 0$$

$$f(a) = 0$$

$$\text{Here } LHL = RHL = f(x) \text{ at } x = a$$

Hence $f(x)$ is continuous at $x = a$.

$$8. f(x) = \begin{cases} \frac{e^x}{1+e^x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases} \text{ at } x = 0.$$

$$\text{Ans: Given } f(x) = \begin{cases} \frac{e^x}{1+e^x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases} \text{ at } x = 0$$

$$LHL = f(x)$$

$$= \lim_{x \rightarrow 0^-} \frac{e^{\frac{1}{x}}}{1+e^{\frac{1}{x}}};$$

$$x \rightarrow 0^- \Rightarrow x < 0 \therefore \frac{1}{x} \rightarrow -\infty$$

$$e^{\frac{1}{x}} \rightarrow 0$$

$$= \lim_{x \rightarrow 0^-} \left(\frac{0}{1+0} \right)$$

$$LHL = 0$$

$$RHL = f(x) = \lim_{x \rightarrow 0^+} \frac{e^{\frac{1}{x}}}{1+e^{\frac{1}{x}}};$$

$$x \rightarrow 0^+ \Rightarrow x > 0 \therefore \frac{1}{x} \rightarrow \infty$$

$$e^{\frac{1}{x}} \rightarrow \infty$$

$$= \lim_{x \rightarrow 0^+} \frac{1}{1 + \frac{1}{e^x}}$$

$$= \left(\frac{1}{1+0} \right)$$

$$RHL = 1$$

$$f(0) = 0$$

Here $LHL \neq RHL$ at $x = 0$

Hence $f(x)$ is discontinuous at $x = 0$.

$$9. f(x) = \begin{cases} \frac{x^2}{2}, & \text{if } 0 \leq x \leq 1 \\ 2x^2 - 3x + \frac{3}{2}, & \text{if } 1 < x \leq 2 \end{cases} \text{ at } x = 1.$$

$$\text{Ans: Given } f(x) = \begin{cases} \frac{x^2}{2}, & \text{if } 0 \leq x \leq 1 \\ 2x^2 - 3x + \frac{3}{2}, & \text{if } 1 < x \leq 2 \end{cases} \text{ at } x = 1$$

$$LHL = f(x)$$

$$= \lim_{x \rightarrow 1^-} \frac{x^2}{2}$$

$$= \lim_{x \rightarrow 1^-} \left(\frac{1}{2} \right)$$

$$LHL = \frac{1}{2}$$

$$RHL = f(x)$$

$$= \lim_{x \rightarrow 1^+} \left(2x^2 - 3x + \frac{3}{2} \right)$$

$$= \lim_{x \rightarrow 1^+} \left(2 \times 1^2 - 3 \times 1 + \frac{3}{2} \right)$$

$$= \left(\frac{3}{2} - 1 \right)$$

$$RHL = \frac{1}{2}$$

$$f(1) = \frac{1}{2}$$

Here $LHL = RHL = f(1)$ at $x = 1$

Hence $f(x)$ is continuous at $x = 1$.

10. $f(x) = |x| + |x-1|$ at $x = 1$.

Ans:

Given $f(x) = |x| + |x-1|$ at $x = 1$

$$LHL = f(x)$$

$$= \lim_{x \rightarrow 1^-} |x| + |x-1|$$

$$\text{as } x < 1 \therefore |x| = x, |x-1| = -(x-1)$$

$$= \lim_{x \rightarrow 1^-} (x - x + 1)$$

$$LHL = 1$$

$$RHL = f(x)$$

$$= \lim_{x \rightarrow 1^+} |x| + |x-1|$$

$$\text{as } x > 1 \therefore |x| = x, |x-1| = (x-1)$$

$$= \lim_{x \rightarrow 1^+} (2x - 1)$$

$$= (2 - 1)$$

$$RHL = 1$$

$$f(1) = 1$$

Here $LHL = RHL = f(1)$ at $x = 1$

Hence $f(x)$ is continuous at $x = 1$.

Find the value of k in each of the Exercises 11 to 14 so that the function f is continuous at the indicated point:

11. $f(x) = \begin{cases} 3x - 8, & \text{if } x \leq 5 \\ 2k, & \text{if } x > 5 \end{cases}$ at $x = 5$.

Ans:

Given $f(x) = \begin{cases} 3x - 8, & \text{if } x \leq 5 \\ 2k, & \text{if } x > 5 \end{cases}$ at $x = 5$

$f(x)$ will be continuous at $x = 5$ if

$$LHL = RHL = f(5)$$

$$LHL = f(x)$$

$$= \lim_{x \rightarrow 5^-} (3x - 8)$$

$$= \lim_{x \rightarrow 5^-} (15 - 8)$$

$$LHL = 7$$

$$RHL = f(x)$$

$$= \lim_{x \rightarrow 5^+} (2k)$$

$$RHL = 2k$$

$$2k = 7 \Rightarrow k = \frac{7}{2}$$

12. $f(x) = \begin{cases} \frac{2^{x+2} - 16}{4^x - 16}, & \text{if } x \neq 2 \\ 2k, & \text{if } x = 2 \end{cases}$ at $x = 2$.

Ans:

Given $f(x) = \begin{cases} \frac{2^{x+2} - 16}{4^x - 16}, & \text{if } x \neq 2 \\ k, & \text{if } x = 2 \end{cases}$ at $x = 2$

$f(x)$ will be continuous at $x = 2$ if

$$LHL = RHL = f(2)$$

$$LHL = f(x)$$

$$= \lim_{x \rightarrow 2^-} \frac{2^{x+2} - 16}{4^x - 16}$$

$$= \lim_{x \rightarrow 2^-} \frac{4(2^x - 4)}{(2^x - 4)(2^x + 4)}$$

$$= \lim_{x \rightarrow 2^-} \left(\frac{4}{2^2 + 4} \right)$$

$$LHL = \frac{1}{2}$$

$$RHL = f(x)$$

$$= \lim_{x \rightarrow 2^+} \frac{2^{x+2} - 16}{4^x - 16}$$

$$= \lim_{x \rightarrow 2^+} \frac{4(2^x - 4)}{(2^x - 4)(2^x + 4)}$$

$$= \lim_{x \rightarrow 2^+} \left(\frac{4}{2^2 + 4} \right)$$

$$RHL = \frac{1}{2}$$

$$k = \frac{1}{2}$$

$$13. f(x) = \begin{cases} \frac{\sqrt{1+kx} - \sqrt{1-kx}}{x}, & \text{if } -1 \leq x < 0 \\ \frac{2x+1}{x-1}, & \text{if } 0 \leq x \leq 1 \end{cases} \text{ at } x = 0.$$

Ans:

$$\text{Given } f(x) = \begin{cases} \frac{\sqrt{1+kx} - \sqrt{1-kx}}{x}, & \text{if } -1 \leq x < 0 \\ \frac{2x+1}{x-1}, & \text{if } 0 \leq x \leq 1 \end{cases} \text{ at } x=0$$

$f(x)$ will be continuous at $x=0$ if

$$LHL = RHL = f(0)$$

$$LHL = f(x)$$

$$= \lim_{x \rightarrow 0^-} \left(\frac{\sqrt{1+kx} - \sqrt{1-kx}}{x} \right)$$

$$= \lim_{x \rightarrow 0^-} \left(\frac{\sqrt{1+kx} - \sqrt{1-kx}}{x} \right) \left(\frac{\sqrt{1+kx} + \sqrt{1-kx}}{\sqrt{1+kx} + \sqrt{1-kx}} \right)$$

$$= \lim_{x \rightarrow 0^-} \frac{(1+kx) - (1-kx)}{x(\sqrt{1+kx} + \sqrt{1-kx})}$$

$$= \lim_{x \rightarrow 0^-} \frac{2kx}{x(\sqrt{1+kx} + \sqrt{1-kx})}$$

$$C. = \lim_{x \rightarrow 0^-} \frac{2k}{(\sqrt{1+0} + \sqrt{1-0})}$$

$$LHL = k$$

$$RHL = f(x)$$

$$= \lim_{x \rightarrow 0^+} \left(\frac{2x+1}{x-1} \right)$$

$$RHL = -1$$

$$k = -1$$

$$14. f(x) = \begin{cases} \frac{1 - \cos kx}{x \sin x}, & \text{if } x \neq 0 \\ \frac{1}{2}, & \text{if } x = 0 \end{cases} \text{ at } x=0.$$

Ans:

Given $f(x) = \begin{cases} \frac{1 - \cos kx}{x \sin x}, & \text{if } x \neq 0 \\ \frac{1}{2}, & \text{if } x = 0 \end{cases}$ at $x = 0$

$f(x)$ will be continuous at $x = 0$ if

$$LHL = RHL = f(0)$$

$$LHL = RHL = f(x)$$

$$= \lim_{x \rightarrow 0^-} \left(\frac{1 - \cos kx}{x \sin x} \right)$$

$$= \lim_{x \rightarrow 0^-} \left(\frac{1 - \cos kx}{k^2 x^2 \left(\frac{\sin x}{x} \right)} \right) k^2$$

$$= \lim_{x \rightarrow 0^-} \left(\frac{1}{2} \right) k^2$$

$$LHL = RHL = \frac{k^2}{2}$$

$$f(0) = \frac{1}{2}$$

$$\frac{k^2}{2} = \frac{1}{2} \Rightarrow k^2 = 1$$

$$k = \pm 1.$$

15. Prove that the function f defined by $f(x) = \begin{cases} \frac{x}{|x| + 2x^2}, & \text{if } x \neq 0 \\ k, & \text{if } x = 0 \end{cases}$ remains discontinuous at $x = 0$, regardless the choice of k .

Ans:

Given that, $f(x) = \begin{cases} \frac{x}{|x| + 2x^2}, & \text{if } x \neq 0 \\ k, & \text{if } x = 0 \end{cases}$ here to check continuity at $x = 0$

$$LHL = \lim_{x \rightarrow 0^-} f(x)$$

$$= \frac{x}{|x| + 2x^2}$$

$$|x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$$

$$= \lim_{x \rightarrow 0^-} \frac{x}{-x + 2x^2}$$

$$= \lim_{x \rightarrow 0^-} \frac{1}{-1 + 2x}$$

$$LHL = -1$$

$$RHL = \lim_{x \rightarrow 0^+} f(x)$$

$$= \frac{x}{|x| + 2x^2}$$

$$|x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$$

$$= \lim_{x \rightarrow 0^+} \frac{x}{x + 2x^2}$$

$$= \lim_{x \rightarrow 0^+} \frac{1}{1 + 2x}$$

$$RHL = 1$$

$$\text{re } LHL \neq RHL$$

Hence limit doesn't exist, the discontinuity is irremovable so there is no point of value of function at $x = 0$ for continuity. Proved.

16. Find the values of a and b such that the function f defined by

$$f(x) = \begin{cases} \frac{x-4}{|x-4|} + a, & \text{if } x < 4 \\ a + b, & \text{if } x = 4 \\ \frac{x-4}{|x-4|} + b, & \text{if } x > 4 \end{cases} \text{ is a continuous function}$$

at $x = 4$.

Ans:

Given that $f(x) = \begin{cases} \frac{x-4}{|x-4|} + a, & \text{if } x < 4 \\ a+b, & \text{if } x = 4 \\ \frac{x-4}{|x-4|} + b, & \text{if } x > 4 \end{cases}$

We must find a and b such that function be continuous at $x = 4$.

$$LHL = \lim_{x \rightarrow 4^-} f(x)$$

$$= \frac{x-4}{|x-4|} + a$$

$$|x-4| = \begin{cases} -(x-4), & x < 4 \\ x-4, & x \geq 4 \end{cases}$$

$$= \lim_{x \rightarrow 4^-} \frac{x-4}{-(x-4)} + a$$

$$= \lim_{x \rightarrow 4^-} (-1 + a)$$

$$LHL = a - 1$$

$$RHL = \lim_{x \rightarrow 4^+} f(x) = \frac{x-4}{|x-4|} + b$$

$$|x-4| = \begin{cases} -(x-4), & x < 4 \\ x-4, & x \geq 4 \end{cases}$$

$$= \lim_{x \rightarrow 4^+} \frac{x-4}{(x-4)} + b$$

$$= \lim_{x \rightarrow 4^+} (1 + b)$$

$$RHL = b + 1$$

$$f(4) = a + b$$

$$LHL = RHL = f(4)$$

$$a - 1 = b + 1 = a + b$$

$$a = 1, b = -1$$

17. Given the function $f(x) = \frac{1}{x+2}$. Find the points of discontinuity of the composite function $y = f(f(x))$.

Ans:

Given that $f(x) = \frac{1}{x+2}$,

$$\text{Now, } f(f(x)) = f\left(\frac{1}{x+2}\right) = \frac{1}{\frac{1}{x+2} + 2} = \frac{x+2}{1+2x+4} = \frac{x+2}{2x+5}$$

Here $y = \frac{x+2}{2x+5}$ so, for y to be defined $2x+5 \neq 0$

$$2x \neq -5 \Rightarrow x \neq -\frac{5}{2}$$

Hence $f(f(x))$ is discontinuous at $x = -\frac{5}{2}$.

18. Find all points of discontinuity of the function $f(t) = \frac{1}{t^2+t-2}$, where $t = \frac{1}{x-1}$.

Ans:

Given, $f(t) = \frac{1}{t^2+t-2}$ and $t = \frac{1}{x-1}$

$$f(x) = \frac{1}{\left(\frac{1}{x-1}\right)^2 + \left(\frac{1}{x-1}\right) - 2}$$

$$= \frac{(x-1)^2}{1 + (x-1) - 2(x-1)^2}$$

$$= \frac{(x-1)^2}{-2(x^2 - 2x + 1) + x - 1 + 1}$$

$$= \frac{(x-1)^2}{-2x^2 + 5x - 2}$$

For $f(x)$ to be defined $-2x^2 + 5x - 2 \neq 0$

$$-(2x^2 - 4x - x + 2) \neq 0$$

$$2x(x-2) - (x-2) \neq 0$$

$$(2x-1)(x-2) \neq 0$$

$$x \neq \frac{1}{2} \text{ and } 2$$

Hence points of discontinuity for the function $f(x)$ are

$$x = \frac{1}{2} \text{ and } x = 2.$$

19. Show that the function $f(x) = |\sin \sin x + \cos \cos x|$ is continuous at $x = \pi$.

Ans:

Given $f(x) = |\sin \sin x + \cos \cos x|$ to check continuity at $x = \pi$.

$$f(x) = |\sin \sin x + \cos \cos x|$$

$$= |\sin \sin \pi + \cos \cos \pi|$$

$$= |0 - 1|$$

$$f(x) = 1$$

$$f(\pi) = |\sin \sin \pi + \cos \cos \pi| = |0 - 1| = 1$$

So, $f(x)$ is continuous at $x = \pi$.

Hence proved.

Examine the differentiability of function f , where f is defined by

20. $f(x) = \{x[x], \text{ if } 0 \leq x < 2(x-1)x, 2 \leq x < 3 \text{ at } x = 2.$

Ans:

Given $f(x) = \{x[x], \text{ if } 0 \leq x < 2(x-1)x, 2 \leq x < 3 \text{ at } x = 2,$

$$LHD = f'(2^-)$$

$$= \frac{f(2-h) - f(2)}{-h}$$

$$= \frac{(2-h)[2-h] - 2(2-1)}{-h}$$

$$= \frac{(2-h)1 - 2}{-h}$$

$$= \frac{-h}{-h}$$

$$LHD = 1$$

$$RHD = f'(2^+)$$

$$= \frac{f(2+h) - f(2)}{h}$$

$$= \frac{(2+h-1)(2+h) - 2(2-1)}{h}$$

$$= \frac{h^2 + 3h + 2 - 2}{h}$$

$$= \frac{h^2 + 3h}{h}$$

$$RHD = 3$$

Here $LHD \neq RHD$

Hence function f is not differentiable at $x = 2$.

21. $f(x) = \begin{cases} x^2 \cdot \sin \frac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{at } x = 0 \end{cases}$

Ans:

Given $f(x) = \begin{cases} x^2 \cdot \sin \frac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{at } x = 0 \end{cases}$

$$LHD = f'(0^-)$$

$$= \frac{f(0-h) - f(0)}{-h}$$

$$= \frac{(0-h)^2 \sin \left(\frac{1}{0-h} \right) - 0}{-h}$$

$$= \frac{-h^2 \sin \left(\frac{1}{h} \right)}{-h}$$

$$= h \sin \left(\frac{1}{h} \right) ; \because 0 \times (\text{numbers vary between } -1 \text{ to } 1)$$

$$LHD = 0$$

$$RHD = f'(0^+)$$

$$= \frac{f(0+h) - f(0)}{h}$$

$$= \frac{(0+h)^2 \sin \left(\frac{1}{0+h} \right) - 0}{h}$$

$$= \frac{h^2 \sin \left(\frac{1}{h} \right)}{h}$$

$$= h \sin \left(\frac{1}{h} \right) ; \because 0 \times (\text{numbers vary between } -1 \text{ to } 1)$$

$$RHD = 0 \text{ Here } LHD = RHD$$

Hence function f is differentiable at $x = 0$.

22. $f(x) = \begin{cases} 1+x, & \text{if } x \leq 2 \\ 5-x, & \text{if } x > 2 \end{cases}$ at $x = 2$.

Ans:

Given $f(x) = \begin{cases} 1+x, & \text{if } x \leq 2 \\ 5-x, & \text{if } x > 2 \end{cases}$ at $x = 2$,

$$LHD = f'(2^-)$$

$$= \frac{f(2-h) - f(2)}{-h}$$

$$= \frac{(1+2-h) - 3}{-h}$$

$$= \frac{-h}{-h}$$

$$= 1$$

$$LHD = 1$$

$$RHD = f'(2^+)$$

$$= \frac{f(2+h) - f(2)}{h}$$

$$= \frac{(5-2-h) - 3}{h}$$

$$= \frac{-h}{h}$$

$$= -1$$

$$RHD = -1$$

Here $LHD \neq RHD$

Hence function f is not differentiable at $x = 2$.

23. Show that $f(x) = |x - 5|$ is continuous but not differentiable at $x = 5$.

Ans:

Given that $f(x) = |x - 5|$

Here, $f(x) = |x - 5| = \begin{cases} -(x - 5), & x < 5 \\ x - 5, & x \geq 5 \end{cases}$

Continuity at $x = 5$

$$LHL = f(x)$$

$$= \lim_{x \rightarrow 5^-} (5 - x)$$

$$= 0$$

$$RHL = f(x)$$

$$= \lim_{x \rightarrow 5^+} (x - 5)$$

$$= 0$$

$$f(5) = 0$$

Here $LHL = RHL = f(5)$

Hence f is continuous at $x = 5$.

Differentiability at $x = 5$

$$LHD = f'(5^-)$$

$$= \lim_{h \rightarrow 0} \left(\frac{1}{-h} \right) (f(5 - h) - f(5))$$

$$= \lim_{h \rightarrow 0} \left(\frac{1}{-h} \right) \{|5 - h - 5| - 0\}$$

$$= \lim_{h \rightarrow 0} \left(\frac{1}{-h} \right) h$$

$$LHD = -1$$

$$RHD = f'(5^+)$$

$$= \lim_{h \rightarrow 0} \left(\frac{1}{h} \right) (f(5+h) - f(5))$$

$$= \lim_{h \rightarrow 0} \left(\frac{1}{h} \right) \{ |5+h-5| - 0 \}$$

$$= \lim_{h \rightarrow 0} \left(\frac{1}{h} \right) h$$

$$RHD = 1$$

Here, $LHD \neq RHD$

Hence f is not differentiable at $x = 5$.

24.A function $f : R \rightarrow R$ satisfies the equation $f(x+y) = f(x) \cdot f(y)$ for all $x, y \in R, f(x) \neq 0$. Suppose that the function is differentiable at $x = 0$ and $f'(0) = 2$, then prove that $f'(x) = 2f(x)$.

Ans:

Given $f : R \rightarrow R$ satisfies the equation $f(x+y) = f(x) \cdot f(y)$.

for all $x, y \in R, f(x) \neq 0$.

$$\begin{aligned} f'(x) &= \frac{1}{h} \{ f(x+h) - f(x) \} \\ &= \frac{1}{h} \{ f(x) \cdot f(h) - f(x) \}; \because f(x+y) = f(x) \cdot f(y) \\ &= f(x) \frac{1}{h} \{ f(h) - 1 \} \end{aligned}$$

By putting $x = 0 = y$, $f(x+y) = f(x) \cdot f(y)$

$$f(0) = f(0) \cdot f(0) \Rightarrow f(0) = 0 \text{ or } 1 \text{ but } f(x) \neq 0$$

So, $f(0) = 1$

$$f'(x) = f(x) \frac{1}{h} \{ f(h) - f(0) \}$$

$$= f(x) \cdot f'(0)$$

$$f'(x) = 2f(x)$$

Hence proved.

Differentiate each of the following w.r.to x (Exercise 25 to 43)

25. 2^x

Ans:

Given $y = 2^x$

By taking log to the base e .

$$\ln \ln y = \ln \ln 2^x = x \ln \ln 2$$

By differentiating w.r.to x we have

$$\frac{d}{dx} \ln \ln y = \ln \ln 2 \cdot \frac{d}{dx} x$$

$$\frac{d}{dy} \ln \ln y \cdot \frac{dy}{dx} = \ln \ln 2 \cdot \frac{d}{dx} x \cdot \frac{d}{dx} \cos \cos x$$

$$\frac{1}{y} \frac{dy}{dx} = (\ln \ln 2) \cdot (x) \cdot (-\sin \sin x)$$

$$\frac{dy}{dx} = -y \cdot \sin \sin 2x \cdot \ln \ln 2$$

$$\frac{dy}{dx} = -2^x \cdot \sin \sin 2x \cdot \ln \ln 2$$

26. $\frac{8^x}{x^8}$

Ans:

Given $y = \frac{8^x}{x^8}$

By taking log to base e .

$$\ln \ln y = \ln \ln \frac{8^x}{x^8} = \ln \ln 8^x - \ln \ln x^8$$

$$\ln \ln y = x \ln \ln 8 - 8 \ln \ln x$$

$$\frac{d}{dx} \ln \ln y = \frac{d}{dx} (x \ln \ln 8 - 8 \ln \ln x)$$

$$\frac{d}{dy} \ln \ln y \frac{dy}{dx} = \left(\ln \ln 8 - \frac{8}{x} \right)$$

$$\frac{dy}{dx} = y \left(\ln \ln 8 - \frac{8}{x} \right)$$

$$\frac{dy}{dx} = \frac{8^x}{x^8} \left(\ln \ln 8 - \frac{8}{x} \right)$$

27. $\log \log (x + \sqrt{x^2 + a})$

Ans:

Given that $y = \log \log (x + \sqrt{x^2 + a})$

$$\frac{dy}{dx} = \frac{d}{dx} \left(\log \log (x + \sqrt{x^2 + a}) \right)$$

$$= \frac{d}{d(x + \sqrt{x^2 + a})} \log \log (x + \sqrt{x^2 + a}) \frac{d}{dx} (x + \sqrt{x^2 + a})$$

$$= \frac{1}{(x + \sqrt{x^2 + a})} \left\{ 1 + \frac{d}{d(x^2 + a)} \sqrt{x^2 + a} \frac{d(x^2 + a)}{dx} \right\}$$

$$= \frac{1}{(x + \sqrt{x^2 + a})} \left\{ 1 + \frac{1}{2\sqrt{x^2 + a}} 2x \right\}$$

$$= \frac{1}{\left(x + \sqrt{x^2 + a}\right)} \left\{ \frac{x + \sqrt{x^2 + a}}{\sqrt{x^2 + a}} \right\}$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{x^2 + a}}$$

28. $\log\log\left[\log\log\left(\log\log x^5\right)\right]$

Ans:

Given $y = \log\log\left[\log\log\left(\log\log x^5\right)\right]$

$$\frac{dy}{dx} = \frac{d}{dx} \log\log\left[\log\log\left(\log\log x^5\right)\right]$$

$$= \frac{d}{d\log\log\left(\log\log x^5\right)} \log\log\left[\log\log\left(\log\log x^5\right)\right]$$

$$\frac{d}{d\log\log x^5} \log\log\left(\log\log x^5\right) 5 \frac{d}{dx} \log\log x$$

$$= \frac{1}{\log\log\left(\log\log x^5\right)} \times \frac{1}{\log\log x^5} \times \frac{5}{x}$$

$$\frac{dy}{dx} = \frac{1}{x \cdot \log\log x^5 \cdot \log\log\left(\log\log x^5\right)}$$

29. $\sin\sin\sqrt{x} + \sqrt{x}$

Ans:

Given $y = \sin\sin\sqrt{x} + \sqrt{x}$

$$\frac{dy}{dx} = \frac{d}{dx} \left(\sin\sin\sqrt{x} + \sqrt{x} \right)$$

$$\begin{aligned}
 &= \frac{d}{d\sqrt{x}} \sin \sin \sqrt{x} \frac{d\sqrt{x}}{dx} + \frac{d}{d\sqrt{x}} \sqrt{x} \frac{d\sqrt{x}}{dx} \\
 &= \cos \cos \sqrt{x} \frac{1}{2\sqrt{x}} - \sin \sin 2\sqrt{x} \frac{1}{2\sqrt{x}} \\
 &= \frac{1}{2\sqrt{x}} (\cos \cos \sqrt{x} - \sin \sin 2\sqrt{x})
 \end{aligned}$$

30. $(ax^2 + bx + c)$

Ans:

Given $y = (ax^2 + bx + c)$

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d}{dx} (ax^2 + bx + c) \\
 &= \frac{d(ax^2 + bx + c)}{d \sin(ax^2 + bx + c)} \times \frac{d \sin(ax^2 + bx + c)}{d(ax^2 + bx + c)} \times \frac{d(ax^2 + bx + c)}{dx} \\
 &= n(ax^2 + bx + c) \cdot \cos \cos(ax^2 + bx + c) \cdot (2ax + b) \\
 \frac{dy}{dx} &= n(2ax + b) \cdot (ax^2 + bx + c) \cdot \cos \cos(ax^2 + bx + c)
 \end{aligned}$$

31. $\cos \cos(\tan \sqrt{x+1})$

Ans:

Given $y = \cos \cos(\tan \sqrt{x+1})$

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d}{dx} \cos \cos(\tan \sqrt{x+1}) \\
 &= \frac{d \cos \cos(\tan \sqrt{x+1})}{d(\tan \sqrt{x+1})} \times \frac{d(\tan \sqrt{x+1})}{d\sqrt{x+1}} \times \frac{d\sqrt{x+1}}{dx}
 \end{aligned}$$

$$= -\sin \sin \left(\tan \tan \sqrt{x+1} \right) \cdot \sqrt{x+1} \cdot \frac{1}{2\sqrt{x+1}}$$

$$\frac{dy}{dx} = \frac{-1}{2\sqrt{x+1}} \cdot \sin \sin \left(\tan \tan \sqrt{x+1} \right) \cdot \sqrt{x+1}$$

32. $\sin \sin x^2 + x + x^2$

Ans:

Given $y = \sin \sin x^2 + x + x^2$

$$\frac{dy}{dx} = \frac{d}{dx} \left(\sin \sin x^2 + x + x^2 \right)$$

$$= \frac{d \sin \sin x^2}{dx^2} \times \frac{dx^2}{dx} + \frac{dx}{dx} + \frac{dx^2}{dx^2} \times \frac{dx^2}{dx}$$

$$\frac{dy}{dx} = 2x \cos \cos x^2 + \sin \sin 2x + 2x \sin \sin (2x^2)$$

33. $\left(\frac{1}{\sqrt{x+1}} \right)$

Ans:

Given $y = \left(\frac{1}{\sqrt{x+1}} \right)$

$$\frac{dy}{dx} = \frac{d}{dx} \left\{ \left(\frac{1}{\sqrt{x+1}} \right) \right\}$$

$$= \frac{d \left(\frac{1}{\sqrt{x+1}} \right)}{d \left(\frac{1}{\sqrt{x+1}} \right)} \times \frac{d \left(\frac{1}{\sqrt{x+1}} \right)}{d \sqrt{x+1}} \times \frac{d \sqrt{x+1}}{dx}$$

$$= \frac{1}{\sqrt{1 - \left(\frac{1}{\sqrt{x+1}}\right)^2}} \times \left\{ -\frac{1}{\left(\sqrt{x+1}\right)^2} \right\} \times \frac{1}{2\sqrt{x+1}}$$

$$= -\frac{\sqrt{x+1}}{\sqrt{x+1}-1} \times \frac{1}{\left(\sqrt{x+1}\right)^2} \times \frac{1}{2\sqrt{x+1}}$$

$$\frac{dy}{dx} = -\frac{1}{2\sqrt{x}(x+1)}$$

34. $(\sin \sin x)^{\cos \cos x}$

Ans:

Given $y = (\sin \sin x)^{\cos \cos x}$

Taking log

$$\log \log y = \log \log \left((\sin \sin x)^{\cos \cos x} \right) = \cos \cos x \log \log \sin \sin x$$

$$\frac{d}{dx} \log \log y = \frac{d}{dx} \cos \cos x \log \log \sin \sin x$$

$$\frac{d}{dy} \log \log y \cdot \frac{dy}{dx} = \cos \cos x \frac{d \log \log \sin \sin x}{d \sin \sin x} \times \frac{d \sin \sin x}{dx} + \log \log \sin \sin x \cdot \frac{d \cos \cos x}{dx}$$

$$\frac{1}{y} \frac{dy}{dx} = \cos \cos x \cdot \frac{1}{\sin \sin x} \cdot \cos \cos x - \sin \sin x \log \log \sin \sin x$$

$$\frac{dy}{dx} = (\sin \sin x)^{\cos \cos x} (\cos \cos x \cdot \cot \cot x - \sin \sin x \log \log \sin \sin x)$$

35. $x \cdot x$

Ans:

Given $y = x \cdot x$

$$\frac{dy}{dx} = \frac{d}{dx}(x \cdot x)$$

$$= x \frac{d}{dx} x + x \frac{d}{dx} x$$

$$= x \cdot mx \cos x + x \cdot nx (-\sin x)$$

$$= mx^2 - nx^2$$

$$= x^2(mx - nx)$$

$$36. (x+1)^2(x+2)^3(x+3)^4$$

Ans:

$$\text{Given } y = (x+1)^2(x+2)^3(x+3)^4$$

$$\frac{dy}{dx} = \frac{d}{dx} \{ (x+1)^2(x+2)^3(x+3)^4 \}$$

$$= (x+2)^3(x+3)^4 \frac{d}{dx}(x+1)^2 + (x+1)^2(x+3)^4 \frac{d}{dx}(x+2)^3 + (x+1)^2(x+2)^3 \frac{d}{dx}(x+3)^4$$

$$= (x+2)^3(x+3)^4 \cdot 2(x+1) + (x+1)^2(x+3)^4 \cdot 3(x+2)^2 + (x+1)^2(x+2)^3 \cdot 4(x+3)^3$$

$$= (x+1)(x+2)^2(x+3)^3 \{ 2(x+2)(x+3) + 3(x+1)(x+3) + 4(x+1)(x+2) \}$$

$$= (x+1)(x+2)^2(x+3)^3 \{ (2+3+4)x^2 + (10+12+12)x + 12+9+8 \}$$

$$= (x+1)(x+2)^2(x+3)^3(9x^2 + 34x + 29)$$

$$37. \left(\frac{\sin x \cos x + \cos x \sin x}{\sqrt{2}} \right), -\frac{\pi}{4} < x < \frac{\pi}{4}$$

Ans:

$$\text{Given } y = \left(\frac{\sin x \cos x + \cos x \sin x}{\sqrt{2}} \right), -\frac{\pi}{4} < x < \frac{\pi}{4}$$

$$= \left(\sin \sin \frac{\pi}{4} \sin \sin x + \cos \cos \frac{\pi}{4} \cos \cos x \right)$$

$$= \left(\cos \cos \left(\frac{\pi}{4} - x \right) \right)$$

$$y = \frac{\pi}{4} - x$$

$$\frac{dy}{dx} = -1$$

$$38. \left(\sqrt{\frac{1 - \cos \cos x}{1 + \cos \cos x}} \right), -\frac{\pi}{4} < x < \frac{\pi}{4}$$

Ans:

$$\text{Given } y = \left(\sqrt{\frac{1 - \cos \cos x}{1 + \cos \cos x}} \right), -\frac{\pi}{4} < x < \frac{\pi}{4}$$

$$= \left(\sqrt{\frac{2^{\frac{x}{2}}}{2^{\frac{x}{2}}}} \right)$$

$$= \left(\sqrt{\frac{x}{2}} \right)$$

$$= \left| \tan \frac{x}{2} \right|$$

$$y = \frac{x}{2}$$

$$\frac{dy}{dx} = \frac{1}{2}$$

39. $(\sec \sec x + \tan \tan x), -\frac{\pi}{2} < x < \frac{\pi}{2}$

Ans:

Given $y = (\sec \sec x + \tan \tan x), -\frac{\pi}{2} < x < \frac{\pi}{2}$

$$= \frac{(1 + \sin \sin x)}{\cos \cos x}$$

$$= \frac{1 - \cos \cos \left(\frac{\pi}{2} + x \right)}{\sin \sin \left(\frac{\pi}{2} + x \right)}$$

$$= \frac{2 \left(\frac{\pi}{4} + \frac{x}{2} \right)}{2 \sin \sin \left(\frac{\pi}{4} + \frac{x}{2} \right) \cos \cos \left(\frac{\pi}{4} + \frac{x}{2} \right)}$$

$$= \left(\tan \tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right)$$

$$y = \frac{\pi}{4} + \frac{x}{2}$$

$$\frac{dy}{dx} = \frac{1}{2}$$

40. $\left(\frac{a \cos \cos x - b \sin \sin x}{b \cos \cos x + a \sin \sin x} \right), -\frac{\pi}{2} < x < \frac{\pi}{2} \text{ and } \frac{a}{b} \tan \tan x \rangle - 1$

Ans:

Given $y = \left(\frac{a \cos \cos x - b \sin \sin x}{b \cos \cos x + a \sin \sin x} \right), -\frac{\pi}{2} < x < \frac{\pi}{2} \text{ and } \frac{a}{b} \tan \tan x \rangle - 1$

$$= \left(\frac{a \cos \cos x}{b \cos \cos x} - \frac{b \sin \sin x}{a \sin \sin x} \right)$$

$$= \left(\frac{\frac{a}{b} - \tan \tan x}{1 + \frac{a}{b} \tan \tan x} \right)$$

$$= \left(\frac{a}{b} \right) - x$$

$$\frac{dy}{dx} = -1$$

41. $\left(\frac{1}{4x^3 - 3x} \right), 0 < x < \frac{1}{\sqrt{2}}$

Ans:

Given $y = \left(\frac{1}{4x^3 - 3x} \right), 0 < x < \frac{1}{\sqrt{2}}$

Let, $x = \cos \cos t$

$$y = \left(\frac{1}{4t - 3 \cos \cos t} \right)$$

$$= \left(\frac{1}{\cos \cos 3t} \right)$$

$$= (\sec \sec 3t)$$

$$= 3t$$

$$y = 3x$$

$$\frac{dy}{dx} = 3 \times \frac{-1}{\sqrt{1-x^2}} = \frac{-3}{\sqrt{1-x^2}}$$

$$42. \left(\frac{3a^2x - x^3}{a^3 - 3ax^2} \right), -\frac{1}{\sqrt{3}} < \frac{x}{a} < \frac{1}{\sqrt{3}}$$

Ans:

$$\text{Given } y = \left(\frac{3a^2x - x^3}{a^3 - 3ax^2} \right), -\frac{1}{\sqrt{3}} < \frac{x}{a} < \frac{1}{\sqrt{3}}$$

$$y = \left(\frac{3 \cdot \frac{x}{a} - \left(\frac{x}{a} \right)^3}{1 - 3 \left(\frac{x}{a} \right)^2} \right)$$

$$\text{Let, } \frac{x}{a} = \tan t$$

$$y = (\tan 3t); \because \tan 3t = \frac{3 \tan t - t}{1 - 3t}$$

$$y = 3t = 3 \cdot \frac{x}{a}$$

$$\frac{dy}{dx} = 3 \cdot \frac{1}{a \left(1 + \left(\frac{x}{a} \right)^2 \right)}$$

$$\frac{dy}{dx} = \frac{3a}{a^2 + x^2}$$

$$43. \left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right), -1 < x < 1, x \neq 0$$

Ans:

$$\text{Given } y = \left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right), -1 < x < 1, x \neq 0$$

$$\text{Let, } x^2 = \cos 2t$$

$$y = \left(\frac{\sqrt{1 + \cos 2t} + \sqrt{1 - \cos 2t}}{\sqrt{1 + \cos 2t} - \sqrt{1 - \cos 2t}} \right)$$

$$= \left(\frac{\sqrt{2} \cos t + \sqrt{2} \sin t}{\sqrt{2} \cos t - \sqrt{2} \sin t} \right)$$

$$= \left(\frac{1 + \tan t}{1 - \tan t} \right)$$

$$= 1 + (\tan t) = \frac{\pi}{4} + t$$

$$= \frac{\pi}{4} + \frac{1}{2} x^2$$

$$\frac{dy}{dx} = \frac{1}{2} \times \frac{-2x}{\sqrt{1-x^4}}$$

$$\frac{dy}{dx} = \frac{-x}{\sqrt{1-x^4}}$$

Find $\frac{dy}{dx}$ of each of the functions expressed in parametric form in Exercises from 44 to 48.

44. $x = t + \frac{1}{t}, y = t - \frac{1}{t}$

Ans:

Given $x = t + \frac{1}{t}, y = t - \frac{1}{t}$

$$\frac{dx}{dt} = \frac{d}{dt} \left(t + \frac{1}{t} \right), \frac{dy}{dt} = \frac{d}{dt} \left(t - \frac{1}{t} \right)$$

$$\frac{dx}{dt} = 1 - \frac{1}{t^2}, \frac{dy}{dt} = 1 + \frac{1}{t^2}$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

$$= \frac{1 + \frac{1}{t^2}}{1 - \frac{1}{t^2}}$$

$$= \frac{t^2 + 1}{t^2 - 1}$$

$$x + y = 2t \Rightarrow t = \frac{x + y}{2}$$

$$\frac{dy}{dx} = \frac{(x + y)^2 + 4}{(x + y)^2 - 4}$$

45. $x = e^\theta \left(\theta + \frac{1}{\theta} \right), y = e^\theta \left(\theta - \frac{1}{\theta} \right)$

Ans:

Given $x = e^\theta \left(\theta + \frac{1}{\theta} \right), y = e^\theta \left(\theta - \frac{1}{\theta} \right)$

$$\frac{dx}{d\theta} = \frac{d}{d\theta} \left\{ e^\theta \left(\theta + \frac{1}{\theta} \right) \right\}, \frac{dy}{d\theta} = \frac{d}{d\theta} \left\{ e^\theta \left(\theta - \frac{1}{\theta} \right) \right\}$$

$$\frac{dx}{d\theta} = e^\theta \left(\theta + \frac{1}{\theta} \right) + e^\theta \left(1 - \frac{1}{\theta^2} \right), \frac{dy}{d\theta} = -e^\theta \left(\theta - \frac{1}{\theta} \right) + e^\theta \left(1 + \frac{1}{\theta^2} \right)$$

$$\frac{dy}{dx} = \frac{dy / d\theta}{dx / d\theta}$$

$$= \frac{-e^\theta \left(\theta - \frac{1}{\theta} \right) + e^\theta \left(1 + \frac{1}{\theta^2} \right)}{e^\theta \left(\theta + \frac{1}{\theta} \right) + e^\theta \left(1 - \frac{1}{\theta^2} \right)}$$

$$= e^{-2\theta} \left(\frac{\frac{-(\theta^2 - 1)}{\theta} + \frac{(\theta^2 + 1)}{\theta^2}}{\frac{(\theta^2 + 1)}{\theta} + \frac{(\theta^2 - 1)}{\theta^2}} \right)$$

$$\frac{dy}{dx} = e^{-2\theta} \left(\frac{-\theta^3 + \theta^2 + \theta + 1}{\theta^3 + \theta^2 + \theta - 1} \right)$$

46. $x = 3\cos\theta\cos\theta - 2\theta$, $y = 3\sin\theta\sin\theta - 2\theta$

Ans:

Given $x = 3\cos\theta\cos\theta - 2\theta$, $y = 3\sin\theta\sin\theta - 2\theta$

$$\frac{d}{d\theta} x = \frac{d}{d\theta} (3\cos\theta\cos\theta - 2\theta), \frac{d}{d\theta} y = \frac{d}{d\theta} (3\sin\theta\sin\theta - 2\theta)$$

$$\frac{dx}{d\theta} = -3\sin\theta\sin\theta - 6\theta (-\sin\theta), \frac{dy}{d\theta} = 3\cos\theta\cos\theta - 6\theta \cos\theta$$

$$\frac{dy}{dx} = \frac{dy / d\theta}{dx / d\theta}$$

$$\frac{dy}{dx} = \frac{3\cos\theta\cos\theta - 6\theta \cos\theta}{-3\sin\theta\sin\theta + 6\sin\theta\theta}$$

$$= \frac{3\cos\theta\cos\theta}{3\sin\theta\sin\theta} \times \frac{1 - 2\theta}{2\theta - 1}$$

$$\frac{dy}{dx} = \cot\theta\cot\theta$$

47. $\sin x = \frac{2t}{1+t^2}$, $\tan y = \frac{2t}{1-t^2}$

Ans:

Given $\sin x = \frac{2t}{1+t^2}$, $\tan y = \frac{2t}{1-t^2}$

Let $t = \tan\theta$

$$\sin x = \frac{2\tan\theta}{1+\theta} , \tan y = \frac{2\tan\theta}{1-\theta}$$

$$\sin x = \sin 2\theta , \tan y = \tan 2\theta$$

$$x = 2\theta, y = 2\theta$$

$$y = x$$

$$\frac{dy}{dx} = 1$$

48. $x = \frac{1 + \log \log t}{t^2}, y = \frac{3 + 2 \log \log t}{t}.$

Ans:

Given that $x = \frac{1 + \log \log t}{t^2}, y = \frac{3 + 2 \log \log t}{t}$

$$\frac{dx}{dt} = \frac{d}{dt} \left(\frac{1 + \log \log t}{t^2} \right), \frac{dy}{dt} = \frac{d}{dt} \left(\frac{3 + 2 \log \log t}{t} \right)$$

$$\frac{dx}{dt} = \frac{t^2 \left(\frac{1}{t} \right) - (1 + \log \log t) 2t}{t^4}, \frac{dy}{dt} = \frac{t \left(\frac{2}{t} \right) - (3 + 2 \log \log t)}{t^2}$$

$$\frac{dx}{dt} = -\frac{1 + 2 \log \log t}{t^3}, \frac{dy}{dt} = -\frac{1 + 2 \log \log t}{t^2}$$

$$\frac{dy}{dx} = \frac{dy / dt}{dx / dt}$$

$$\frac{dy}{dx} = \left(-\frac{1 + 2 \log \log t}{t^2} \right) \times \left(-\frac{t^3}{1 + 2 \log \log t} \right)$$

$$\frac{dy}{dx} = t$$

49. If $x = e^{\cos 2t}$ and $y = e^{\sin 2t}$, then prove that $\frac{dy}{dx} = -\frac{y \log x}{x \log y}.$

Ans:

Given that $x = e^{\cos 2t} \Rightarrow \cos 2t = \log x$

$$y = e^{\sin 2t} \Rightarrow \sin 2t = \log y$$

Squaring and adding we have

$$2t + 2t = (\log x)^2 + (\log y)^2$$

$$(\log x)^2 + (\log y)^2 = 1$$

By differentiating w.r.to x we have

$$\frac{d}{dx}(\log x)^2 + \frac{d}{dx}(\log y)^2 = \frac{d}{dx}1$$

$$\frac{2\log x}{x} + \frac{2\log y}{y} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{y \log x}{x \log y}$$

50.If $x = a \sin 2t (1 + \cos 2t)$ **and** $y = b \cos 2t (1 - \cos 2t)$, **show that**

$$\left(\frac{dy}{dx} \right)_{at t = \frac{\pi}{4}} = \frac{b}{a}.$$

Ans:

Given $x = a \sin 2t (1 + \cos 2t)$ and $y = b \cos 2t (1 - \cos 2t)$

$$x = a(\sin 2t + \sin 2t \cos 2t), y = b(\cos 2t - \cos 2t \cos 2t)$$

$$x = a\left(\sin 2t + \frac{1}{2} \sin 4t\right), y = b\left(\cos 2t - \frac{1}{2} \cos 4t\right)$$

$$\frac{dx}{dt} = a(2\cos 2t + 2\cos 4t), \frac{dy}{dt} = b(-2\sin 2t + 2\sin 4t)$$

$$\frac{dy}{dx} = \frac{b(-2\sin 2t + 2\sin 4t)}{a(2\cos 2t + 2\cos 4t)}$$

$$At t = \frac{\pi}{4}$$

$$\left(\frac{dy}{dx}\right)_{at t=\frac{\pi}{4}} = \frac{b\left(-\sin\sin\frac{\pi}{2} + \sin\sin\pi\right)}{a\left(\cos\cos\frac{\pi}{2} + \cos\cos\pi\right)}$$

$$\left(\frac{dy}{dx}\right)_{at t=\frac{\pi}{4}} = \frac{b(-1)}{a(-1)} = \frac{b}{a}$$

Hence proved.

51.If $x = 3\sin\sin t - \sin\sin 3t$, $y = 3\cos\cos t - \cos\cos 3t$, **find** $\frac{dy}{dx}$ at $t = \frac{\pi}{3}$.

Ans:

Given $x = 3\sin\sin t - \sin\sin 3t$, $y = 3\cos\cos t - \cos\cos 3t$

$x = 4t$, $y = 3\cos\cos t - \cos\cos 3t$

$$\frac{dx}{dt} = 12t \cos\cos t, \frac{dy}{dt} = -3\sin\sin t + 3\sin\sin 3t$$

$$\frac{dy}{dx} = \frac{-3\sin\sin t + 3\sin\sin 3t}{12t \cos\cos t}$$

$$at t = \frac{\pi}{3}$$

$$\left(\frac{dy}{dx}\right)_{at t=\frac{\pi}{3}} = \frac{-\sin\sin\frac{\pi}{3} + \sin\sin\pi}{4\frac{\pi}{3}\cos\cos\frac{\pi}{3}} = \frac{-\frac{\sqrt{3}}{2}}{4 \times \left(\frac{\sqrt{3}}{2}\right)^2 \times \frac{1}{2}}$$

$$\left(\frac{dy}{dx}\right)_{at t=\frac{\pi}{3}} = -\frac{1}{\sqrt{3}}$$

52.Differentiate $\frac{x}{\sin\sin x}$ **w.r.to** $\sin\sin x$.

Ans:

Given functions are $f(x) = \frac{x}{\sin \sin x}$, $g(x) = \sin \sin x$

Differentiation of f w.r.to g

$$\frac{df(x)}{dg(x)} = \frac{df(x)/dx}{dg(x)/dx}$$

$$\frac{df(x)}{dx} = \frac{d}{dx} \left(\frac{x}{\sin \sin x} \right) = \frac{\sin \sin x - x \cos \cos x}{x}$$

$$\frac{dg(x)}{dx} = \frac{d}{dx} \sin \sin x = \cos \cos x$$

$$\frac{df(x)}{dg(x)} = \left(\frac{\sin \sin x - x \cos \cos x}{x} \right) \left(\frac{1}{\cos \cos x} \right)$$

$$\frac{df(x)}{dg(x)} = \frac{\tan \tan x - x}{x}$$

53. Differentiate $\left(\frac{\sqrt{1+x^2}-1}{x} \right)$ w.r.to x when $x \neq 0$.

Ans:

Given functions are $f(x) = \left(\frac{\sqrt{1+x^2}-1}{x} \right)$, $g(x) = x$

Differentiation of f w.r.to g

$$\frac{df(x)}{dg(x)} = \frac{df(x)/dx}{dg(x)/dx}$$

$$\frac{df(x)}{dx} = \frac{d}{dx} \left(\frac{\sqrt{1+x^2}-1}{x} \right), \frac{dg(x)}{dx} = \frac{d}{dx} x$$

For f , let $x = \tan t$

$$\left(\frac{\sqrt{1+t} - 1}{\tan t} \right) = \left(\frac{(\sec t - 1) \cos t}{\sin t} \right)$$

$$= \left(\frac{1 - \cos t}{\sin t} \right)$$

$$= \left(\tan \frac{t}{2} \right)$$

$$= \frac{t}{2} = \frac{1}{2} x$$

$$\frac{df(x)}{dx} = \frac{d}{dx} \left(\frac{\sqrt{1+x^2} - 1}{x} \right)$$

$$= \frac{d}{dx} \left(\frac{1}{2} x \right)$$

$$= \frac{1}{2(1+x^2)}$$

$$\frac{dg(x)}{dx} = \frac{d}{dx} x = \frac{1}{1+x^2}$$

$$\frac{df(x)}{dg(x)} = \frac{df(x)/dx}{dg(x)/dx} = \frac{\frac{1}{2(1+x^2)}}{\frac{1}{1+x^2}} = \frac{1}{2}$$

Find $\frac{dy}{dx}$ when x and y are connected by the relation given in each of the exercise

54 to 57.

54. $\sin(xy) + \frac{x}{y} = x^2 - y$

Ans:

Given $\sin \sin(xy) + \frac{x}{y} = x^2 - y$

Differentiating w.r.to x we have

$$\frac{d}{dx} \sin \sin(xy) + \frac{d}{dx} \left(\frac{x}{y} \right) = \frac{d}{dx} x^2 - \frac{dy}{dx}$$

$$\cos \cos(xy) \left\{ y + x \frac{dy}{dx} \right\} + \frac{y - x \frac{dy}{dx}}{y^2} = 2x - \frac{dy}{dx}$$

$$y^3 \cos \cos(xy) + \cos \cos(xy) y^2 x \frac{dy}{dx} + y - x \frac{dy}{dx} = 2xy^2 - y^2 \frac{dy}{dx}$$

$$(\cos \cos(xy) y^2 x - x + y^2) \frac{dy}{dx} = 2xy^2 - y - y^3 \cos \cos(xy)$$

$$\frac{dy}{dx} = \frac{2xy^2 - y - y^3 \cos \cos(xy)}{\cos \cos(xy) y^2 x - x + y^2}$$

55. $\sec \sec(x+y) = xy$

Ans:

Given $\sec \sec(x+y) = xy$

$$\frac{d}{dx} \sec \sec(x+y) = \frac{d}{dx} (xy)$$

$$\sec \sec(x+y) \tan \tan(x+y) \left\{ 1 + \frac{dy}{dx} \right\} = y + x \frac{dy}{dx}$$

$$\sec \sec(x+y) \tan \tan(x+y) + \sec \sec(x+y) \tan \tan(x+y) \frac{dy}{dx} = y + x \frac{dy}{dx}$$

$$\sec \sec(x+y) \tan \tan(x+y) \frac{dy}{dx} - x \frac{dy}{dx} = y - \sec \sec(x+y) \tan \tan(x+y)$$

$$\frac{dy}{dx} = \frac{y - \sec \sec(x+y) \tan \tan(x+y)}{\sec \sec(x+y) \tan \tan(x+y) - x}$$

$$56. (x^2 + y^2) = a.$$

Ans:

$$\text{Given } (x^2 + y^2) = a$$

$$\therefore x^2 + y^2 = \tan a$$

Differentiating w.r.to x we get

$$2x + 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

$$57. (x^2 + y^2)^2 = xy.$$

Ans:

$$\text{Given } (x^2 + y^2)^2 = xy$$

$$x^4 + y^4 + 2x^2y^2 - xy = 0$$

Differentiating w.r.to x we have

$$4x^3 + 4y^3 \frac{dy}{dx} + 4xy^2 + 4x^2y \frac{dy}{dx} - y - x \frac{dy}{dx} = 0$$

$$(4y^3 + 4x^2y - x) \frac{dy}{dx} = y - 4xy^2 - 4x^3$$

$$\frac{dy}{dx} = \frac{4y^3 + 4x^2y - x}{4y^3 + 4x^2y - x}$$

$$58. \text{If } ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0, \text{ then show that } \frac{dy}{dx} \square \frac{dx}{dy} = 1.$$

Ans:

Given $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

Differentiating w.r.to x we have

$$2ax + 2h\left(y + x\frac{dy}{dx}\right) + 2by\frac{dy}{dx} + 2g + 2f\frac{dy}{dx} = 0$$

$$(hx + by + f)\frac{dy}{dx} = -(ax + hy + g)$$

$$\frac{dy}{dx} = -\frac{ax + hy + g}{hx + by + f}$$

Now differentiating w.r.to y we have

$$2ax\frac{dx}{dy} + 2h\left(x + y\frac{dx}{dy}\right) + 2by + 2g\frac{dx}{dy} + 2f = 0$$

$$(ax + hy + g)\frac{dx}{dy} = -(2hx + by + f)$$

$$\frac{dx}{dy} = -\frac{hx + by + f}{ax + hy + g}$$

Now, $\frac{dy}{dx} \cdot \frac{dx}{dy} = 1$.

Hence proved.

59.If $x = e^{\frac{x}{y}}$, prove that $\frac{dy}{dx} = \frac{x-y}{x \log x}$

Ans:

Given that $x = e^{\frac{x}{y}}$

Taking log, we have

$$\log x = \frac{x}{y}$$

$$y \log \log x = x$$

Differentiating w.r.to x we have

$$\frac{y}{x} + \log \log x \frac{dy}{dx} = 1$$

$$\log \log x \frac{dy}{dx} = 1 - \frac{y}{x} = \frac{x-y}{x}$$

$$\frac{dy}{dx} = \frac{x-y}{x \log \log x}$$

Hence proved.

60. If $y^x = e^{y-x}$, prove that $\frac{dy}{dx} = \frac{(1 + \log \log y)^2}{\log \log y}$.

Ans:

Given $y^x = e^{y-x}$

Taking log, we have

$$x \log \log y = y - x$$

Differentiating w.r.to x we have

$$\log \log y + \frac{x}{y} \frac{dy}{dx} = \frac{dy}{dx} - 1$$

$$1 + \log \log y = \left(1 - \frac{x}{y}\right) \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{y(1 + \log \log y)}{y - x}$$

Now $x \log \log y = y - x$

$$\log \log y = \frac{y}{x} - 1$$

$$1 + \log \log y = \frac{y}{x}$$

$$1 - \frac{x}{y} = 1 - \frac{1}{1 + \log \log y} = \frac{1 + \log \log y - 1}{1 + \log \log y} = \frac{\log \log y}{1 + \log \log y}$$

$$\frac{dy}{dx} = \frac{(1 + \log \log y)}{1 - \frac{x}{y}} = \frac{(1 + \log \log y)}{\frac{\log \log y}{1 + \log \log y}}$$

$$\frac{dy}{dx} = \frac{(1 + \log \log y)^2}{\log \log y}$$

Hence proved.

61.If $y = (\cos \cos x)^{(\cos \cos x)^x \dots \infty}$, **show that** $\frac{dy}{dx} = \frac{y^2 \tan x}{y \log \cos \cos x - 1}$.

Ans:

Given $y = (\cos \cos x)^{(\cos \cos x)^x \dots \infty}$

$$y = (\cos \cos x)^y$$

$$\log \log y = y \log \log \cos \cos x$$

$$\frac{d}{dx} \log \log y = \frac{d}{dx} (y \log \log \cos \cos x)$$

$$\frac{1}{y} \frac{dy}{dx} = \log \log \cos \cos x \frac{dy}{dx} + y \frac{1}{\cos \cos x} (-\sin \sin x)$$

$$\frac{dy}{dx} = y \log \log \cos \cos x \frac{dy}{dx} - y^2 \tan x$$

$$y^2 \tan x = (y \log \log \cos \cos x - 1) \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{y^2 \tan x}{y \log \log \cos \cos x - 1}$$

Hence proved.

62.If $x \sin(a+y) + \sin a \cos(a+y) = 0$, **prove that** $\frac{dy}{dx} = \frac{(a+y)}{\sin a}$.

Ans:

$$\text{Given } x \sin(a+y) + \sin a \cos(a+y) = 0$$

$$\frac{d}{dx}(x \sin(a+y) + \sin a \cos(a+y)) = 0$$

$$\sin(a+y) + x \cos(a+y) \frac{dy}{dx} - \sin a \sin(a+y) \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{\sin(a+y)}{\sin a \sin(a+y) - x \cos(a+y)}$$

$$x = -\frac{\sin a \cos(a+y)}{\sin(a+y)}$$

$$\frac{dy}{dx} = \frac{\sin(a+y)}{\sin a \sin(a+y) - \left(-\frac{\sin a \cos(a+y)}{\sin(a+y)} \right) \cos(a+y)}$$

$$= \frac{(a+y)}{\sin a \{(a+y) + (a+y)\}}$$

$$\frac{dy}{dx} = \frac{(a+y)}{\sin a}$$

Hence proved.

63.If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$, **prove that** $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$.

Ans:

$$\text{Given } \sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$$

Let $x = \sin p$, $y = \sin q$

$$1 - x^2 = 1 - \sin^2 p = \cos^2 p$$

$$1 - y^2 = 1 - \sin^2 q = \cos^2 q$$

$$\cos p + \cos q = a(\sin p - \sin q)$$

$$2\cos\frac{p+q}{2}\cos\frac{p-q}{2} = a\left(2\cos\frac{p+q}{2}\sin\frac{p-q}{2}\right)$$

$$\cot\frac{p-q}{2} = a$$

$$p - q = 2a$$

$$x - y = 2a$$

$$\frac{d}{dx}(x - y) = 2 \frac{d}{dx}a$$

$$\frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-y^2}} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$$

Hence prove.

64.If $y = x$, find $\frac{d^2y}{dx^2}$ in terms of y alone.

Ans:

Given $y = x$

$$x = \tan y$$

Differentiating w.r.to x

$$1 = y \frac{dy}{dx}$$

$$\frac{dy}{dx} = y$$

$$\text{Again, } \frac{d}{dx} \frac{dy}{dx} = \frac{d}{dx} y$$

$$\frac{d^2 y}{dx^2} = -\sin \sin 2y \frac{dy}{dx} = -\sin \sin 2y y$$

Verify the Rolle's theorem for each of the functions in Exercise 65 to 69.

65. $f(x) = x(x-1)^2$ in $[0,1]$.

Ans:

$$\text{Given } f(x) = x(x-1)^2$$

Function f is polynomial, so

i. it is continuous and

ii. differentiable for all real x .

$$\text{iii. } f(0) = 0 \text{ and } f(1) = 0$$

All three conditions for Rolle's theorem are verified, so

$$\exists c \in [0,1] \text{ such that } f'(c) = 0.$$

$$f'(x) = (x-1)^2 + 2x(x-1)$$

$$= 3x^2 - 4x + 1$$

$$3c^2 - 4c + 1 = 0$$

$$3c^2 - 3c - c + 1 = 0$$

$$3c(c-1) - (c-1) = 0$$

$$(3c-1)(c-1) = 0$$

$$c = 1, \frac{1}{3}$$

$$\frac{1}{3} \in [0, 1]$$

Hence Rolle's theorem is verified.

66. $f(x) = x + \sin x$ in $\left[0, \frac{\pi}{2}\right]$.

Ans:

Given $f(x) = x + \sin x$

Function f is sine and cosine function, so

i. it is continuous and

ii. differentiable for all real x .

iii. $f(0) = 1$ and $f\left(\frac{\pi}{2}\right) = 1$

All three conditions for Rolle's theorem are verified, so

$$\exists c \in \left[0, \frac{\pi}{2}\right] \text{ such that } f'(c) = 0.$$

$$f'(x) = \frac{d}{dx}(x + \sin x)$$

$$= 1 + \cos x$$

$$= 1 + \cos x$$

$$= 1 + \cos 2c$$

$$1 + \cos 2c = 0$$

$$\cos 2c = -1 \Rightarrow 2c = \pi$$

$$c = 0, \frac{\pi}{2}$$

$$\cos 2c = 0 \Rightarrow 2c = \frac{\pi}{2}$$

$$c = \frac{\pi}{4} \in \left[0, \frac{\pi}{2}\right]$$

Hence Rolle's theorem is verified.

67. $f(x) = \log \log(x^2 + 2) - \log \log 3$ in $[-1, 1]$

Ans:

Given $f(x) = \log \log(x^2 + 2) - \log \log 3$

Function f is logarithmic function and $x^2 + 2 > 0 \forall x \in R$, so

i. it is continuous and

ii. differentiable for all real x .

iii. $f(-1) = 0$ and $f(1) = 0$

All three conditions for Rolle's theorem are verified, so

$\exists c \in [-1, 1]$ such that $f'(c) = 0$.

$$f'(x) = \frac{d}{dx} (\log \log(x^2 + 2) - \log \log 3)$$

$$= \frac{2x}{x^2 + 2}$$

$$\frac{2c}{c^2 + 1} = 0 \Rightarrow c = 0 \in [-1, 1]$$

Hence Rolle's theorem is verified.

68. $f(x) = x(x+3)e^{\frac{x}{2}}$ in $[-3, 0]$.

Ans:

Given $f(x) = x(x+3)e^{\frac{x}{2}}$

Function f is polynomial with exponential function, so

- i. it is continuous and
- ii. differentiable for all real x .
- iii. $f(-3) = 0$ and $f(0) = 0$

All three conditions for Rolle's theorem are verified, so

$\exists c \in [-3, 0]$ such that $f'(c) = 0$.

$$\begin{aligned} f'(x) &= \frac{d}{dx} \left(x(x+3)e^{-\frac{x}{2}} \right) \\ &= (x+3)e^{-\frac{x}{2}} + xe^{-\frac{x}{2}} - \frac{1}{2}x(x+3)e^{-\frac{x}{2}} \\ &= \frac{1}{2}(x^2 + 7x + 6)e^{-\frac{x}{2}} \end{aligned}$$

$$f'(c) = \frac{1}{2}(c^2 + 7c + 6)e^{-\frac{c}{2}} = 0$$

$$\therefore c^2 + c + 6c + 6 = 0$$

$$c(c+1) + 6(c+1) = 0$$

$$(c+1)(c+6) = 0$$

$$c = -6, -1$$

$$c = -1 \in [-3, 0]$$

Hence Rolle's theorem is verified.

69. $f(x) = \sqrt{4-x^2}$ in $[-2, 2]$

Ans:

Given $f(x) = \sqrt{4-x^2}$

Function f is defined $\forall x \in [-2, 2]$, so

- i. it is continuous in $[-2, 2]$ and

ii. differentiable in $(-2, 2)$.

iii. $f(-2) = 0$ and $f(2) = 0$

All three conditions for Rolle's theorem are verified, so

$\exists c \in [-2, 2]$ such that $f'(c) = 0$.

$$f'(x) = \frac{d}{dx} \sqrt{4-x^2}$$

$$= -\frac{x}{\sqrt{4-x^2}}$$

$$f'(c) = -\frac{c}{\sqrt{4-c^2}} = 0$$

$$c = 0 \in (-2, 2)$$

Hence Rolle's theorem is verified.

70. Discuss the applicability of Rolle's theorem on the function given by

$$f(x) = \begin{cases} x^2 + 1, & 0 \leq x \leq 1 \\ 3 - x, & 1 \leq x \leq 2 \end{cases}$$

Ans:

$$\text{Given } f(x) = \begin{cases} x^2 + 1, & 0 \leq x \leq 1 \\ 3 - x, & 1 \leq x \leq 2 \end{cases}$$

At $x = 1$

$$LHL = f(x)$$

$$= \lim_{x \rightarrow 1^-} (x^2 + 1)$$

$$= 1 + 1 = 2$$

$$RHL = \lim_{x \rightarrow 1^+} f(x)$$

$$= \lim_{x \rightarrow 1^+} (3 - x)$$

$$= 3 - 1 = 2$$

$$LHL = RHL = f(1)$$

So, function is continuous.

$$LHD = f'(1^-) = (2x)_{at\ x=1} = 2$$

$$RHD = f'(1^+) = (-x)_{at\ x=1} = -1$$

$LHD \neq RHD$. So, function is not differentiable.

Hence Rolle's theorem is not applicable in $[0, 2]$.

71. Find the points on the curve $y = \cos \cos x - 1$ in $[0, 2\pi]$, where the tangents are parallel to x-axis.

Ans:

Given $y = \cos \cos x - 1$

Tangent is parallel to x-axis then derivative must be zero at those points. So,

$$\frac{dy}{dx} = \frac{d}{dx}(\cos \cos x - 1) = \sin \sin x = 0$$

$$x = n\pi; n \in \mathbb{Z}$$

But in $(0, 2\pi)$, $x = \pi$

Point is $y = \cos \cos \pi - 1 = -2$

$(\pi, -2)$.

Hence the point is $(\pi, -2)$.

72. Using Rolle's theorem, find the point on the curve $y = x(x - 4)$, $x \in [0, 4]$. Where the tangent is parallel to x-axis.

Ans:

Given $y = x(x - 4)$, $x \in [0, 4]$,

Function f is polynomial, so

- i. it is continuous and
- ii. differentiable for all real x .
- iii. $f(0) = 0$ and $f(4) = 0$

All three conditions for Rolle's theorem are verified, so

$\exists c \in [0, 4]$ such that $f'(c) = 0$.

$$f'(x) = \frac{d}{dx}(x^2 - 4x)$$

$$= 2x - 4$$

$$f'(c) = 2x - 4 = 0$$

$$c = 2 \in (0, 4)$$

At $x = 2$ function has tangent parallel to x-axis.

$$\text{Again, } y = 2(2 - 4) = -4$$

Hence the point is $(2, -4)$.

Verify mean value theorem for each of the functions given Exercises 73 to 76.

$$73. f(x) = \frac{1}{4x-1}, x \in [1, 4]$$

Ans:

$$\text{Given } f(x) = \frac{1}{4x-1}, x \in [1, 4]$$

For mean value theorem function must be continuous and differentiable in the given domain.

Here, function f is discontinuous at $x = \frac{1}{4}$.

Again $f'(x) = -\frac{4}{(4x-1)^2}$, so f is not differentiable at $x = \frac{1}{4}$.

Both conditions are satisfied for mean value theorem.

So, there exists a real number $c \in (1,4)$ such that

$$f'(c) = \frac{f(4) - f(1)}{4 - 1}$$

$$-\frac{4}{(4c-1)^2} = \frac{1}{3} \left(\frac{1}{4 \times 4 - 1} - \frac{1}{4 \times 1 - 1} \right)$$

$$\frac{4}{(4c-1)^2} = -\frac{1}{3} \left(\frac{1}{15} - \frac{1}{3} \right)$$

$$\left(\frac{2}{4c-1} \right)^2 = \frac{1}{3} \left(\frac{5-1}{15} \right) = \frac{4}{45}$$

$$\frac{2}{4c-1} = \pm \frac{2}{3\sqrt{5}}$$

$$4c-1 = \pm 3\sqrt{5}$$

$$c = \frac{1}{4} (1 \pm 3\sqrt{5})$$

$$c = \frac{1}{4} (1 + 3\sqrt{5}) \in (1,4)$$

Hence mean value theorem is verified.

74. $f(x) = x^3 - 2x^2 - x + 3, x \in [0,1]$.

Ans:

Given $f(x) = x^3 - 2x^2 - x + 3, x \in [0,1]$

Since, $f(x)$ is polynomial and polynomial function is continuous and differentiable for all real x .

So there exists a real number $c \in (0,1)$ such that

$$f'(c) = \frac{f(1) - f(0)}{1 - 0}$$

$$f'(x) = 3x^2 - 4x - 1$$

$$3c^2 - 4c - 1 = (1 - 2 - 1 + 3) - (3)$$

$$3c^2 - 4c - 1 = -2$$

$$3c^2 - 3c - c + 1 = 0$$

$$3c(c-1) - (c-1) = 0$$

$$(3c-1)(c-1) = 0$$

$$c = 1, \frac{1}{3}$$

$$c = \frac{1}{3} \in (0, 1)$$

Hence mean value theorem is verified.

$$75. f(x) = \sin \sin x - \sin \sin 2x, x \in [0, \pi]$$

Ans:

$$\text{Given that } f(x) = \sin \sin x - \sin \sin 2x, x \in [0, \pi]$$

Since $\sin$ function is continuous and differentiable for all real x , so there exists a real number $c \in (0, \pi)$ such that

$$f'(c) = \frac{f(\pi) - f(0)}{\pi - 0}$$

$$f'(x) = \cos \cos x - 2 \cos \cos 2x$$

$$\cos \cos c - 2 \cos \cos 2c = \frac{1}{\pi} (0 - 0) = 0$$

$$\cos \cos c - 2(2c - 1) = 0$$

$$4c - \cos \cos c - 2 = 0$$

$$\cos \cos c = \frac{1 \pm \sqrt{1 + 4 \times 4 \times 2}}{8} = \frac{1 \pm \sqrt{33}}{8}$$

$$c = \left(\frac{1 \pm \sqrt{33}}{8} \right) \in (0, \pi)$$

Hence mean value theorem is verified.

76. $f(x) = \sqrt{25 - x^2}, x \in [1, 5]$.

Ans:

Given $f(x) = \sqrt{25 - x^2}, x \in [1, 5]$

$f(x)$ is continuous when $25 - x^2 \geq 0$

So, $x \in [-5, 5]$

$\therefore f(x)$ is continuous in $[1, 5]$

Now, $f'(x) = -\frac{x}{\sqrt{25 - x^2}}$ so, it is differentiable in $(1, 5)$

It satisfies both the conditions of mean value theorem.

There exists a real number $c \in (1, 5)$ such that

$$f'(c) = \frac{f(5) - f(1)}{5 - 1}$$

$$-\frac{c}{\sqrt{25 - c^2}} = \frac{1}{4} \left(\sqrt{25 - 5^2} - \sqrt{25 - 1^2} \right)$$

$$\frac{c}{\sqrt{25 - c^2}} = -\frac{1}{4} (-2\sqrt{6})$$

Squaring both sides, we have

$$\frac{c^2}{25 - c^2} = \frac{6}{4} = \frac{3}{2}$$

$$2c^2 = 75 - 3c^2$$

$$5c^2 = 75$$

$$c^2 = 15$$

$$c = \pm\sqrt{15}$$

$$c = \sqrt{15} \in (1, 5)$$

Hence, mean value theorem is verified.

77. Find a point on the curve $y = (x-3)^2$, where the tangent is parallel to the chord joining the points $(3, 0)$ and $(4, 1)$.

Ans:

$$\text{Given } y = (x-3)^2$$

Function is polynomial so, it will be continuous and differentiable for all real x .

Mean value theorem is applicable. Points $(3, 0)$ and $(4, 1)$ are on curve so, there exists a real number $c \in (3, 4)$ such that

$$f'(c) = \frac{f(4) - f(3)}{4 - 3}$$

$$2(c-3) = \frac{1-0}{1} = 1$$

$$c-3 = \frac{1}{2}$$

$$c = 3 + \frac{1}{2} = \frac{7}{2}$$

$$y = \left(\frac{7}{2} - 3\right)^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

Hence the point is $\left(\frac{7}{2}, \frac{1}{4}\right)$.

78.Using mean value theorem, prove that there is a point on the curve $y = 2x^2 - 5x + 3$ between the points $A(1,0)$ and $B(2,1)$, where tangent is parallel to chord AB. Also find the point.

Ans:

Given $y = 2x^2 - 5x + 3$ and points are $A(1,0)$ and $B(2,1)$.

y is a polynomial, so it is continuous and differentiable for all real x . Meanvalue theorem is applicable. Points $(1,0)$ and $(2,1)$ are on curve so, there exists a real number $c \in (1,2)$ such that

$$f'(c) = \frac{f(2) - f(1)}{2 - 1}$$

$$4c - 5 = \frac{1 - 0}{1}$$

$$4c = 6 \Rightarrow c = \frac{3}{2} \in (1,2)$$

There is a tangent to the curve at $x = \frac{3}{2}$ where tangent is parallel to the chord AB.

Hence proved.

$$\text{For the point, } y = 2 \times \left(\frac{3}{2}\right)^2 - 5 \times \frac{3}{2} + 3 = \frac{9}{2} - \frac{15}{2} + \frac{6}{2} = 0$$

Point is $\left(\frac{3}{2}, 0\right)$.

Long Answer (L.A.)

79.Find the values of p and q so that $f(x) = \begin{cases} x^2 + 3x + p, & \text{if } x \leq 1 \\ qx + 2, & \text{if } x > 1 \end{cases}$ is differentiable at $x = 1$.

Ans:

Given that $f(x) = \begin{cases} x^2 + 3x + p, & \text{if } x \leq 1 \\ qx + 2, & \text{if } x > 1 \end{cases}$ at $x = 1$.

For differentiability function must be continuous. So, for continuity

$$LHL = f(x)$$

$$= \lim_{x \rightarrow 1^-} (x^2 + 3x + p)$$

$$LHL = 4 + p$$

$$RHL = f(x)$$

$$= \lim_{x \rightarrow 1^+} (qx + 2)$$

$$RHL = q + 2$$

$$f(1) = p + 4$$

$$LHL = RHL = f(1)$$

$$p + 4 = q + 2$$

$$p = q - 2$$

Now for differentiability

$$LHD = f'(1^-) = (2x + 3)_{at\ x=1} = 5$$

$$RHD = f'(1^+) = q$$

For differentiability $LHD = RHD$

$$q = 5$$

$$p = 5 - 2 = 3$$

Hence value of $p = 3$ and $q = 5$.

80.If $x^m y^n = (x + y)^{m+n}$, prove that

$$\text{i. } \frac{dy}{dx} = \frac{y}{x} \text{ and ii. } \frac{d^2y}{dx^2} = 0$$

Ans:

$$\text{Given } x^m y^n = (x + y)^{m+n}$$

i. $x^m y^n = (x + y)^{m+n}$

$$\frac{d}{dx} x^m y^n = \frac{d}{dx} (x + y)^{m+n}$$

$$x^m n y^{n-1} \frac{dy}{dx} + m x^{m-1} y^n = (m+n)(x+y)^{m+n-1} \left(1 + \frac{dy}{dx}\right)$$

$$\left(x^m n y^{n-1} - (m+n)(x+y)^{m+n-1}\right) \frac{dy}{dx} = (m+n)(x+y)^{m+n-1} - m x^{m-1} y^n$$

$$\frac{dy}{dx} = \frac{(m+n)(x+y)^{m+n-1} - m x^{m-1} y^n}{\left(x^m n y^{n-1} - (m+n)(x+y)^{m+n-1}\right)}$$

$$= \frac{x^m y^n (m+n)(x+y)^{-1} - m x^{m-1} y^n}{\left(x^m n y^{n-1} - x^m y^n (m+n)(x+y)^{-1}\right)}$$

$$= \frac{(m+n)(x+y)^{-1} - m x^{-1}}{\left(n y^{-1} - (m+n)(x+y)^{-1}\right)}$$

$$= \frac{x(m+n) - m(x+y)}{(n(x+y) - (m+n)y)} \cdot \frac{y}{x}$$

$$= \frac{mx + nx - mx - my}{nx + ny - my - ny} \cdot \frac{y}{x}$$

$$= \frac{nx - my}{nx - my} \cdot \frac{y}{x}$$

$$\frac{dy}{dx} = \frac{y}{x}$$

Hence proved.

ii. we have $\frac{dy}{dx} = \frac{y}{x}$

$$\frac{d}{dx} \frac{dy}{dx} = \frac{d}{dx} \left(\frac{y}{x}\right)$$

$$\frac{d^2y}{dx^2} = \frac{x \frac{dy}{dx} - y}{x^2} = \frac{x \cdot \frac{y}{x} - y}{x^2}$$

$$\frac{d^2y}{dx^2} = 0$$

Hence proved.

81.If $x = \sin \sin t$ and $y = \sin \sin pt$, **prove that** $(1 - x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} + p^2 y = 0$.

Ans:

Given that $x = \sin \sin t$ and $y = \sin \sin pt$

$$\frac{dx}{dt} = \cos \cos t, \frac{dy}{dt} = p \cos \cos pt$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{p \cos \cos pt}{\cos \cos t}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dt} \left(\frac{p \cos \cos pt}{\cos \cos t} \right) \frac{1}{dx/dt}$$

$$= p \frac{\cos \cos t (-p \sin \sin pt) - \cos \cos pt (-\sin \sin t)}{t} \cdot \frac{1}{\cos \cos t}$$

$$(1 - x^2) \frac{d^2y}{dx^2} = p(1 - t) \frac{\cos \cos pt \sin \sin t - p \sin \sin pt \cos \cos t}{t}$$

$$(1 - x^2) \frac{d^2y}{dx^2} = p(\cos \cos pt \tan \sin t - p \sin \sin pt)$$

$$x \frac{dy}{dx} = \sin \sin t \frac{p \cos \cos pt}{\cos \cos t} = p \cos \cos pt \tan \sin t$$

$$p^2 y = p^2 \sin \sin pt$$

$$LHS = (1 - x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} + p^2 y$$

$$= p(\cos \cos pt \tan t - p \sin \sin pt) - p \cos \cos pt \tan t + p^2 \sin \sin pt$$

$$= 0$$

Hence proved.

82. Find $\frac{dy}{dx}$, if $y = x^{\tan x} + \sqrt{\frac{x^2 + 1}{2}}$

Ans:

Given $y = x^{\tan x} + \sqrt{\frac{x^2 + 1}{2}}$

$$p = x^{\tan x}$$

Taking log,

$$\log \log p = \tan x \log \log x$$

$$\frac{d}{dx} \log \log p = \frac{d}{dx} (\tan x \log \log x)$$

$$\frac{1}{p} \frac{dp}{dx} = \frac{\tan x}{x} + x \log \log x$$

$$\frac{d}{dx} (x^{\tan x}) = x^{\tan x} \left(\frac{\tan x}{x} + x \log \log x \right)$$

$$\frac{d}{dx} \left(\sqrt{\frac{x^2 + 1}{2}} \right) = \frac{1}{\sqrt{2}} \left(\frac{x}{\sqrt{x^2 + 1}} \right)$$

$$\frac{dy}{dx} = \frac{d}{dx} (x^{\tan x}) + \frac{d}{dx} \left(\sqrt{\frac{x^2 + 1}{2}} \right)$$

$$\frac{dy}{dx} = x^{\tan x} \left(\frac{\tan x}{x} + x \log \log x \right) + \frac{1}{\sqrt{2}} \left(\frac{x}{\sqrt{x^2 + 1}} \right)$$

Chose the correct answers from the given four options in each of the Exercises 83 to 96.

83. If $f(x) = 2x$ and $g(x) = \frac{x^2}{2} + 1$, then which of the following can be a discontinuous function

A. $f(x) + g(x)$

D. $f(x) - g(x)$

B. $f(x) \cdot g(x)$

C. $\frac{g(x)}{f(x)}$

Ans:

Given that $f(x) = 2x$ and $g(x) = \frac{x^2}{2} + 1$.

We can observe that both the functions are continuous so,

$f + g$, $f - g$ and $f \cdot g$ are continuous but $\frac{g}{f}$ is not continuous at the point where f is zero.

Hence $\frac{g(x)}{f(x)}$ can be discontinuous. Option (D) is correct.

84. The function $f(x) = \frac{4 - x^2}{4x - x^3}$

A. Discontinuous at only one point

B. Discontinuous at exactly two points

C. Discontinuous at exactly three points

D. None of these

Ans:

Given $f(x) = \frac{4-x^2}{4x-x^3}$

For points of discontinuity $4x-x^3=0$

$$x(4-x^2)=0$$

$$x(2-x)(2+x)=0$$

$$x=0, -2, 2$$

Hence $f(x)$ is discontinuous at exactly three points.

Option (C) is correct.

85. The set of points where the function f given by $f(x) = |2x-1|\sin x$ is differentiable is

A. R B. $R - \left\{\frac{1}{2}\right\}$ C. $(0, \infty)$ D. None of these.

Ans:

Given function is $f(x) = |2x-1|\sin x$

$$f(x) = \begin{cases} -(2x-1)\sin x, & x \leq \frac{1}{2} \\ (2x-1)\sin x, & x \geq \frac{1}{2} \end{cases}$$

We know that modulus function is continuous everywhere but not differentiable at

$$2x-1=0 \Rightarrow x = \frac{1}{2}$$

Hence, $f(x)$ is differentiable in $R - \left\{\frac{1}{2}\right\}$.

86. The function $f(x) = \cot \cot x$ is discontinuous on the set

A. $\{x = n\pi : n \in \mathbb{Z}\}$

E. $\{x = 2n\pi : n \in \mathbb{Z}\}$

$$\text{F. } \left\{ x = (2n+1)\frac{\pi}{2} : n \in \mathbb{Z} \right\}$$

$$\text{G. } \left\{ x = \frac{n\pi}{2} : n \in \mathbb{Z} \right\}$$

Ans:

Given $f(x) = \cot \cot x$

We know that $f(x) = \cot \cot x$ is continuous at $x \in \mathbb{R} - \{n\pi, n \in \mathbb{Z}\}$

Hence, $f(x) = \cot \cot x$ is discontinuous at $\{x = n\pi : n \in \mathbb{Z}\}$.

87. The function $f(x) = e^{|x|}$ is

A. Continuous everywhere but not differentiable at $x = 0$

B. Continuous and differentiable everywhere

C. Not continuous at $x = 0$

D. None of these

Ans:

Given $f(x) = e^{|x|}$

$$f(x) = e^{|x|} = \begin{cases} e^{-x}, & x \leq 0 \\ e^x, & x \geq 0 \end{cases}$$

$$LHL = e^0 = 1$$

$$RHL = e^0 = 1$$

$LHL = RHL = f(0)$, function is continuous.

$$f'(x) = \begin{cases} -e^{-x}, & x < 0 \\ e^x, & x > 0 \end{cases}$$

$$LHD = -e^0 = -1$$

$$RHD = e^0 = 1$$

$LHD \neq RHD$, function is not differentiable.

Hence continuous everywhere but not differentiable at $x = 0$

Option (A) is correct.

88.If $f(x) = x^2 \sin \frac{1}{x}$, where $x \neq 0$, then the value of the function f at $x = 0$, so that the function is continuous at $x = 0$, is

A. 0 B. -1 C. 1 D. None of these

Ans:

Given $f(x) = x^2 \sin \frac{1}{x}$, where $x \neq 0$

$$f(x) = \lim_{x \rightarrow 0} x^2 \sin \frac{1}{x}$$

$$= 0 \times (\text{number varying between } -1 \text{ to } 1)$$

$$= 0$$

Hence value of $f(x)$ at $x = 0$, so that $f(x)$ be continuous at $x = 0$, is 0. Option (A) is correct.

89.If $f(x) = \begin{cases} mx + 1, & \text{if } x \leq \frac{\pi}{2} \\ \sin x + n, & \text{if } x > \frac{\pi}{2} \end{cases}$ is continuous at $x = \frac{\pi}{2}$ then

H. $m = 1, n = 0$

I. $m = \frac{n\pi}{2} + 1$

J. $n = \frac{m\pi}{2}$

K. $m = n = \frac{\pi}{2}$

Ans:

Given $f(x) = \begin{cases} mx + 1, & \text{if } x \leq \frac{\pi}{2} \\ \sin x + n, & \text{if } x > \frac{\pi}{2} \end{cases}$

$$LHL = f(x)$$

$$= \lim_{x \rightarrow \frac{\pi}{2}^-} (mx + 1)$$

$$= \frac{m\pi}{2} + 1$$

$$RHL = \lim_{x \rightarrow \frac{\pi}{2}^+} f(x)$$

$$= \lim_{x \rightarrow \frac{\pi}{2}^+} (\sin \sin x + n)$$

$$= n + 1$$

For $f(x)$ to be continuous

$$LHL = RHL$$

$$\frac{m\pi}{2} + 1 = n + 1$$

$$n = \frac{m\pi}{2}$$

Option (C) is correct.

90. Let $f(x) = |\sin \sin x|$, then

A. f is everywhere differentiable

B. f is everywhere continuous but not differentiable at $x = n\pi, n \in \mathbb{Z}$.

C. f is everywhere continuous but not differentiable at $x = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$.

D. None of these.

Ans:

$$\text{Given } f(x) = |\sin \sin x|$$

$$f(x) = |\sin \sin x| = \begin{cases} -\sin \sin x, & \sin \sin x \leq 0 \\ \sin \sin x, & \sin \sin x \geq 0 \end{cases}$$

Here we observe that f is continuous.

$$f'(x) = \{-\cos \cos x, \sin \sin x \langle 0 \cos \cos x, \sin \sin x \rangle 0\}$$

At $x = n\pi, n \in \mathbb{Z}$

$$LHD = f'(n\pi) = -\cos \cos n\pi = -(-1)^n$$

$$RHD = f'(n\pi) = \cos \cos n\pi = (-1)^n$$

$$LHD \neq RHD$$

Hence f is continuous but not differentiable at $x = n\pi, n \in \mathbb{Z}$.

Option (B) is correct.

91.If $y = \log \log \left(\frac{1-x^2}{1+x^2} \right)$ **then** $\frac{dy}{dx}$ **is equal to**

A. $\frac{4x^3}{1-x^4}$ **B.** $\frac{-4x}{1-x^4}$ **C.** $\frac{1}{4-x^4}$ **D.** $\frac{-4x^3}{1-x^4}$

Ans:

$$\text{Given } y = \log \log \left(\frac{1-x^2}{1+x^2} \right) = \log \log (1-x^2) - \log \log (1+x^2)$$

$$\frac{dy}{dx} = \frac{d}{dx} (\log \log (1-x^2)) - \frac{d}{dx} (\log \log (1+x^2))$$

$$= \frac{-2x}{1-x^2} - \frac{2x}{1+x^2}$$

$$= -2x \left(\frac{1-x^2+1+x^2}{(1-x^2)(1+x^2)} \right)$$

$$= -\frac{4x}{1-x^4}$$

Option (B) is correct.

92.If $y = \sqrt{\sin \sin x + y}$, then $\frac{dy}{dx}$ is equal to

A. $\frac{\cos \cos x}{2y-1}$ B. $\frac{\cos \cos x}{1-2y}$ C. $\frac{\sin \sin x}{1-2y}$ D. $\frac{\sin \sin x}{2y-1}$

Ans:

Given $y = \sqrt{\sin \sin x + y}$

$$y^2 = \sin \sin x + y$$

$$y^2 - y = \sin \sin x$$

$$\frac{d}{dx}(y^2 - y) = \frac{d}{dx} \sin \sin x$$

$$(2y - 1) \frac{dy}{dx} = \cos \cos x$$

$$\frac{dy}{dx} = \frac{\cos \cos x}{2y - 1}$$

Option (A) is correct.

93.The derivative of $(2x^2 - 1)$ w.r.to x .

A. $\frac{-1}{2\sqrt{1-x^2}}$ B. $\frac{2}{x}$ C. $1-x^2$ D. $1-x^2$

Ans:

We must find derivative of $(2x^2 - 1)$ w.r.to x

Let $t = x \Rightarrow x = \cos t$

$$(2x^2 - 1) = (2t - 1)$$

$$= (\cos \cos 2t)$$

$$(2x^2 - 1) = 2t$$

$$\frac{d(2x^2 - 1)}{dx} = \frac{d(2t)}{dt} = 2$$

Option (A) is correct.

94.If $x = t^2, y = t^3$, **then** $\frac{d^2y}{dx^2}$ **is**

A. $\frac{3}{2}$ **B.** $\frac{3}{4t}$ **C.** $\frac{3}{2t}$ **D.** $\frac{3}{4}$

Ans:

Given $x = t^2, y = t^3$

$$\frac{dx}{dt} = 2t, \frac{dy}{dt} = 3t^2$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3t^2}{2t} = \frac{3t}{2}$$

$$\frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{3t}{2} \right) = \frac{d}{dt} \left(\frac{3t}{2} \right) \cdot \frac{1}{dx/dt}$$

$$\frac{d^2y}{dx^2} = \frac{3}{2} \times \frac{1}{2t} = \frac{3}{4t}$$

Option (B) is correct.

95.The value of c in Rolle's theorem for the function $f(x) = x^3 - 3x$ **in the interval** $[0, \sqrt{3}]$ **is**

A. 1 **B.** -1 **C.** $\frac{3}{2}$ **D.** $\frac{1}{3}$

Ans:

Given that $f(x) = x^3 - 3x$

Function is continuous and differentiable for all real x and $f(0) = 0 = f(\sqrt{3})$.

So, Rolle's theorem is applicable here.

$$f'(x) = 3x^2 - 3$$

$$f'(c) = 0 \Rightarrow 3(c^2 - 1) = 0$$

$$c^2 = 1 \Rightarrow c = \pm 1$$

$$c = 1 \in (0, \sqrt{3})$$

Option (A) is correct.

96. For the function $f(x) = x + \frac{1}{x}$, $x \in [1, 3]$, the value of c for mean value theorem is

A. 1 B. $\sqrt{3}$ C. 2 D. None of these

Ans:

Given $f(x) = x + \frac{1}{x}$ is continuous in $x \in [1, 3]$

$f'(x) = 1 - \frac{1}{x^2}$ is differentiable in $x \in (1, 3)$

Mean value theorem is applicable here.

$$f'(c) = 1 - \frac{1}{c^2} = \frac{f(3) - f(1)}{3 - 1}$$

$$1 - \frac{1}{c^2} = \frac{3 + \frac{1}{3} - 1 - 1}{2}$$

$$\frac{1}{c^2} = 1 - \frac{\frac{4}{3}}{2} = \frac{2 - \frac{4}{3}}{2} = \frac{2}{6} = \frac{1}{3}$$

$$c^2 = 3 \Rightarrow c = \pm\sqrt{3}$$

$$c = \sqrt{3} \in (1, 3)$$

Option (B) is correct.

Fill in the blanks in each of the exercises 97 to 101:

97. An example of a function which is continuous everywhere but fails to be differentiable exactly at two points is _____.

Ans:

Function $|x-1| + |x-2|$ is continuous everywhere but it is not differentiable exactly at two points and the points are $x=0, x=1$.

98. Derivative of x^2 w.r.to x^3 is _____.

Ans:

To find derivative of x^2 w.r.to x^3

$$\begin{aligned} \frac{dx^2}{dx^3} &= \frac{\frac{dx^2}{dx}}{\frac{dx^3}{dx}} \\ &= \frac{2x}{3x^2} \\ &= \frac{2}{3x} \end{aligned}$$

99. If $f(x) = |\cos \cos x|$, then $f'\left(\frac{\pi}{4}\right) =$ _____.

Ans:

Given $f(x) = |\cos \cos x|$

At $x = \frac{\pi}{4}$, $f(x) = \cos \cos x$

$$f'(x) = -\sin \sin x$$

$$f'\left(\frac{\pi}{4}\right) = -\sin \sin\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$$

100.If $f(x) = |\cos \cos x - \sin \sin x|$, **then** $f'\left(\frac{\pi}{3}\right) = \underline{\hspace{2cm}}$.

Ans:

Given $f(x) = |\cos \cos x - \sin \sin x|$

$$f(x) = |\cos \cos x - \sin \sin x| = \begin{cases} \sin \sin x - \cos \cos x, & \sin \sin x \geq \cos \cos x \\ -\sin \sin x + \cos \cos x, & \sin \sin x < \cos \cos x \end{cases}$$

At $x = \frac{\pi}{3}, \sin \sin x \geq \cos \cos x$

$$f(x) = \sin \sin x - \cos \cos x$$

$$f'(x) = \cos \cos x + \sin \sin x$$

$$f'\left(\frac{\pi}{3}\right) = \cos \cos \frac{\pi}{3} + \sin \sin \frac{\pi}{3} = \frac{1}{2} + \frac{\sqrt{3}}{2}$$

$$f'\left(\frac{\pi}{3}\right) = \frac{1 + \sqrt{3}}{2}$$

101.For the curve $\sqrt{x} + \sqrt{y} = 1$, $\frac{dy}{dx}$ **at** $\left(\frac{1}{4}, \frac{1}{4}\right)$ **is** $\underline{\hspace{2cm}}$.

Ans:

Given curve is $\sqrt{x} + \sqrt{y} = 1$

$$\frac{d}{dx}(\sqrt{x} + \sqrt{y}) = \frac{d}{dx}1$$

$$\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{\sqrt{y}}{\sqrt{x}}$$

$$\text{At } \left(\frac{1}{4}, \frac{1}{4}\right), \frac{dy}{dx} = -1$$

State True or False for the following statements in each of the Exercise 102 to 106.

102. Rolle's theorem is applicable for the function $f(x) = |x-1|$ in $[0, 2]$.

Ans:

Given function $f(x) = |x-1|$ is continuous in $[0, 2]$ and not differentiable at $x=1 \in (0, 2)$.

Hence given statement is False.

103. If f is continuous on its domain D , then $|f|$ is also continuous on D .

Ans:

The given statement is True.

104. The composition of two continuous functions is a continuous function.

Ans:

The given statement is True.

105. Trigonometric and inverse trigonometric functions are differentiable in their respective domain

Ans:

The given statement is True.

106. If $f \cdot g$ is continuous at $x = a$, then f and g are separately continuous at $x = a$.

Ans:

The given statement is False.

